Treball de Fi de Màster

Master's Degree in Thermal Engineering

Numerical simulation of convection heat transfer problems using CFD & HT techniques

REPORT

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Author: Gonzalo Céspedes Paz

Director: Jesús Castro Gonzàlez

Co-director: Josep Plana Riu

Escola Tècnica Superior
d'Enginyeria Industrial de Barcelona

This document is dedicated to those I had to leave back home to pursue my ambitions; and specially to those I will never get to see again for they are now gone. Your values and ideals will endure time through all the people whose lives you changed during your time. I miss you dearly.

Ini kakiri inauchima umanumuki. Ranautsun upiwata kakiriutsa tanamuki. Tsariwari tamanapan ichari ranatsën.

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Resum

L'objectiu d'aquest projecte és resoldre numèricament les equacions del moviment de fluids amb transferència de calor, és a dir, les equacions de Navier-Stokes acoblades amb l'equació de conservació de l'energia, de manera multidimensional (mètodes CFD i TH). La metodologia es basa en tècniques de volum finit, fent servir el mètode de pas fraccionari per resoldre l'acoblament entre els camps de pressió i velocitat, és a dir, les equacions de conservació de massa i moment. En una etapa posterior, s'incorpora l'equació de conservació de l'energia, permetent l'estudi de problemes de transferència de calor per convecció. L'anàlisi comença amb l'estudi de problemes de transferència de calor per conducció (problema dels 4 materials). En una segona etapa, s'incorpora un camp de velocitat prescrit per estudiar el problema de difusió numèrica en la solució de l'equació de convecció-difusió (problema de Smith-Hutton). En la tercera etapa, es desenvolupa un resoledor del moviment de fluids per a fluxos laminars, és a dir, les equacions de conservació de massa i moment; finalment, s'afegeix l'equació de conservació de l'energia al resoledor de fluids. Els codis computacionals es verifiquen mitjançant estudis de refinament de malla en espai i temps, fent servir casos de referència corresponents a cada etapa. La investigació dels problemes seleccionats inclou estudis paramètrics que consideren variacions en geometria, propietats termofísiques i condicions de contorn.

Resumen

El objetivo de este proyecto es resolver numéricamente las ecuaciones del movimiento de fluidos con transferencia de calor, es decir, las ecuaciones de Navier-Stokes acopladas con la ecuación de conservación de la energía, de manera multidimensional (métodos CFD y TT). La metodología se basa en técnicas de volumen finito, utilizando el método de paso fraccionario para resolver el acoplamiento entre los campos de presión y velocidad, es decir, las ecuaciones de conservación de masa y cantidad de movimiento. En una etapa posterior, se incorpora la ecuación de conservación de la energía, permitiendo el estudio de problemas de transferencia de calor por convección. El análisis comienza con estudios de problemas de transferencia de calor por conducción (problema de los 4 materiales). En una segunda etapa, se incorpora un campo de velocidad prescrito para estudiar el problema de difusión numérica en la solución de la ecuación de convección-difusión (problema de Smith-Hutton). En la tercera etapa, se desarrolla un resolvedor del movimiento de fluidos para flujos laminares, es decir, las ecuaciones de conservación de masa y cantidad de movimiento; finalmente, la ecuación de conservación de la energía se añade al resolvedor de fluidos. Los códigos computacionales se verifican mediante estudios de refinamiento de malla en espacio y tiempo, utilizando casos de referencia correspondientes a cada etapa. La investigación de los problemas seleccionados incluye estudios paramétricos que consideran variaciones en geometría, propiedades termofísicas y condiciones de contorno.

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Abstract

The objective of this project is to numerically solve the equations of fluid motion with heat transfer, i.e., the Navier-Stokes equations coupled with the energy conservation equation, in a multi-dimensional manner (CFD & HT methods). The methodology is based on finite-volume techniques, using the fractional-step method to solve the coupling between the pressure and velocity fields, i.e., the conservation of mass and momentum equations. In a further step, the energy conservation equation is incorporated, enabling the study of convection heat transfer problems. The analysis begins with studies of conduction heat transfer problems (i.e., 4-materials problem). In a second step, a given velocity field is incorporated to study the numerical diffusion problem in the solution of the convection-diffusion equation (i.e., Smith-Hutton problem). In the third step, a fluid motion solver for laminar flows is developed, i.e., the mass and momentum conservation equations; finally, the energy conservation equation is added to the fluid solver. The computational codes are verified through grid-refinement studies in both space and time, using corresponding benchmark cases at each step. The investigation of the selected problems includes parametric studies considering variations in geometry, thermophysical properties, and boundary conditions.

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Introduction

Engineering as a practice has always centered around finding optimized solutions for real-world applications. Throughout history, countless names have devoted their efforts towards the development of mathematical tools and models to further expand the knowledge over physical phenomena. Paired with the rapid increase in computing capacity seen over the last few decades, the current time presents several opportunities to further expand humanity's knowledge base. In the field of Computer Fluid Dynamics and Heat Transfer (CFD & HT), the resolution of the Navier-Stokes equations presents an important challenge that can be optimized for a wide variety of industrial applications. The Navier-Stokes (NS) equations [1] describe the motion of viscous fluids and the complex interactions involved in the phenomena by evaluating the conservation of mass, momentum and energy. However, NS equations are non-linear since the velocity and pressure equations (momentum & mass) in the system are mutually dependent. This adds a layer of complexity for solving the system of equations and in consequence there is no general solution that can be directly solved. However, all conservation equations follow the same physical principle that describes transport phenomena.

This report will cover the development of numerical tools for the resolution of Convection-Diffusion problems while consolidating the knowledge related to the transport phenomena and the application of numerical methods for their resolution. The spatial discretization in this project will follow a finite volume method to discretize the analyzed domain into a finite amount of control volumes. The model will be developed in stages of increasing complexity, starting with pure diffusion, convection-diffusion, and finally coupled NS problems; using well-known benchmarks to validate the obtained results at each stage. During the development, the effect of the different numerical parameters involved in the model will be studied to further understand the mathematical implications of using a numerical tools like the one proposed with this project. Furthermore, the nature of each different case will require the implementation of different numerical solver algorithms that address the optimization problem in different manners.

Finite Volume Method (FVM)

The Finite Volume Method (FVM) discretizes the computational domain into a finite number of non-overlapping *control volumes* (cells) [2, 3]. Rather than solving the governing PDEs in their differential form at isolated points, FVM integrates each equation over every control volume. This integral formulation converts each conservation law into a balance of fluxes across the cell faces, making the method *inherently conservative*: any quantity leaving one cell through a shared face enters the neighbouring cell by exactly the same amount, with no artificial sources or sinks introduced by the discretization.

In this project the domain is divided into a structured rectangular mesh. Each cell is identified by its central node P, and its four neighbours are labelled by cardinal direction: West (W), East (E), South (S), and North (N). Capital letters denote quantities evaluated at the neighbouring cell centroids; lowercase letters denote quantities at the shared face between two cells (west face w, east face e, south face s, north face n). This convention is illustrated in Figure 1. Because all variables are stored at cell centroids, face values must be reconstructed by interpolation from the surrounding centroid values. The choice of interpolation scheme is a central modelling decision, one that directly governs the accuracy and stability of the solver and is discussed in detail throughout this report.

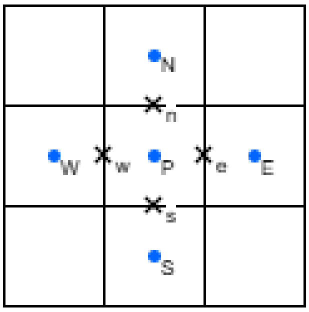

Figure 1: Node layout for a single control volume P. Capital letters (W, E, S, N) denote the four neighbouring cell centroids; lowercase letters (w, e, s, n) denote the corresponding shared faces.

Governing Transport Equations

When studying transport phenomena [1, 4, 5, 2], there are two main mechanisms that govern the propagation of a transport variable ϕ. These are known as **Convection** and **Diffusion**, and classify the phenomena depending on what causes the transport of the transport variable ϕ. Diffusion refers to transport caused by molecular movement within the body, while Convection refers to transport caused by bulk movement of the body. The governing transport equation for this phenomena is given by Eq. (1), which is separated into four terms relating to the physical variation of ϕ. The accumulation term describes the magnitude variation over time, the convective term describes the variation due to convection, the diffusive term describes the variation due to diffusion, and the source term describes any additional source contributions.

$$\overbrace{\frac{\partial(\rho\phi)}{\partial t}}^{\text{Accumulation}} + \overbrace{\nabla(\rho v\phi)}^{\text{Convective}} = \overbrace{\nabla(\Gamma\nabla\phi)}^{\text{Diffusive}} + \overbrace{S_P}^{\text{Source}} \tag{1}$$

The continuous governing equation needs to be integrated over a control volume V_P and a time-step Δt. Applying the Gauss divergence theorem to the volume integrals of the convective and diffusive terms converts them into surface integrals over the cell boundary ∂V_P, as shown in Eq. (2):

$$\int_{V_P} \frac{\partial(\rho\phi)}{\partial t}\, dV + \oint_{\partial V_P} \rho\mathbf{v}\phi \cdot \hat{n}\, dA = \oint_{\partial V_P} \Gamma\nabla\phi \cdot \hat{n}\, dA + \int_{V_P} S_P\, dV \tag{2}$$

Each surface integral can then be approximated as a discrete sum over the four cell faces (west, east, south, north). The resulting discretized forms are shown by Eqs. (3)–(5)

$$\frac{\partial(\rho\phi)}{\partial t} = \rho\frac{\phi_P - \phi_P^0}{\Delta t} \tag{3}$$

The accumulation term follows a first-order implicit approximation, where ϕ_P^0 corresponds to the value of node P at the previous time-step and ϕ_P corresponds to the value at the current time-step being computed.

$$\oint_{\partial V_P} \Gamma \nabla \phi \cdot \hat{n}\, dA = \Gamma_w S_w \frac{\phi_W - \phi_P}{d_{PW}} + \Gamma_e S_e \frac{\phi_E - \phi_P}{d_{PE}} + \Gamma_s S_s \frac{\phi_S - \phi_P}{d_{PS}} + \Gamma_n S_n \frac{\phi_N - \phi_P}{d_{PN}} \tag{4}$$

The diffusive term approximates the gradient at the face by interpolating the values from the adjacent nodes, multiplied by the diffusivity coefficient Γ and the surface at the interface S_k, and dividing by the distance between the central node and one of the adjacent nodes d_{PK}.

$$\oint_{\partial V_P} \rho \mathbf{v} \phi \cdot \hat{n}\, dA = \rho(v_e \phi_e S_e - v_w \phi_w S_w + v_n \phi_n S_n - v_s \phi_s S_s) \tag{5}$$

The convective term uses the face-normal velocities v_k and the face values ϕ_k of the transport variable, each weighted by the corresponding face area S_k. The face values ϕ_k are not directly known and must be interpolated from the adjacent node values; the chosen convective scheme defines this interpolation and considerably influences the accuracy and stability of the model.

By substituting all previously mentioned equations and organizing the different terms, the expression can be reorganized to depend only on the magnitude of the current node P and its neighbours. Eq. (6) shows the standard FVM form for an interior cell with 4 adjacent nodes.

$$a_P \phi_P = a_W \phi_W + a_E \phi_E + a_S \phi_S + a_N \phi_N + b_P \tag{6}$$

The coefficients a_W, a_E, a_S, a_N include the effect of both the diffusive and convective phenomena. When using this discretization, the coefficient for the current cell $a_P = a_W + a_E + a_S + a_N + a_P^0$ will respect diagonal dominance due to the presence of the accumulation term $a_P^0 = \rho V_P / \Delta t$. This equation will be iteratively solved for every node in the discretized domain at each timestep. Each chapter will include a brief explanation of the specific expressions used for each coefficient for their respective application.

All three models in this report are run as transient simulations and advanced in time until the solution reaches a stationary state. Steady state is detected at each time step by monitoring the relative change in the L_2 norm of the primary transported field between successive time steps:

$$\frac{\|\phi^{n+1} - \phi^n\|_2}{\|\phi^n\|_2} < \varepsilon_{\text{temporal}} \tag{7}$$

where ϕ represents the transported variable (T for the diffusion and convection-diffusion models, $\mathbf{v}$ for the Navier-Stokes model), and $\varepsilon_{\text{temporal}}$ is the user-specified temporal tolerance.

Motivation

The study of heat and mass transport phenomena underpins a wide range of engineering applications: thermal management of electronic components, design of heat exchangers and boilers, analysis of combustion chambers, building climatisation, and simulation of geophysical flows. The ability to numerically predict these processes reduces reliance on costly physical prototyping and opens access to design spaces that are impractical to explore experimentally.

Computational Fluid Dynamics and Heat Transfer (CFD & HT) solvers are increasingly central to both research and industrial practice. Developing a working understanding of the underlying numerical methods — rather than treating commercial solvers as black boxes — is essential for critically interpreting simulation results, identifying sources of numerical error, and selecting appropriate physical models. The Navier-Stokes equations are the foundation of any incompressible flow solver, yet their non-linearity and the velocity-pressure coupling they entail present substantial numerical challenges that require careful algorithmic choices.

This project builds a finite volume solver from the ground up in C++, progressing through three stages of increasing physical complexity. This progression simultaneously consolidates knowledge of transport phenomena and develops the numerical reasoning and software engineering skills required to work in modern simulation environments. Each stage is validated against established benchmark cases, closing the loop between theoretical understanding and practical verification.

Objectives

1. Develop a modular FVM solver using C++ capable of accurately simulating transport phenomena of increasing complexity. Using several benchmark cases to validate the accuracy of the implementation at each stage of the model, and testing the performance and behaviour of the different numerical methods implemented in the model once validated.

2. Expand the knowledge about numerical methods applied to CFD & HT and use it to solve and understand the resolution methodology of complex transport phenomena. Simultaneously using the opportunity to further develop a consistent and organized code structure and learning more about C++ and its capabilities as a programming language.

Scope

The model developed in this project will solve three main models, testing a variety of benchmark problems for each one of them.

1. The **pure diffusion** model will evaluate a purely diffusive heat conduction test scenario in 1D and 2D. This first model has the simplest discretization and will serve as a base implementation to test the data structure design for the code implementation. It will also serve as a first introduction to numerical parameters such as the time-step and temporal scheme, and allow for testing to be done regarding their effect on the performance of the model. For the 1D case, the code will be validated against the analytical solution for all implemented boundary combinations in a **1D Rod Case** [5]. For the 2D case, the model will be validated against the **4 Materials Case** benchmark [5]. In addition, this first case will be used to compare the Gauss-Seidel (GS) and Conjugate Gradient (CG) numerical algorithms in terms of performance.

2. The **convection-diffusion** model will evaluate a complete convection-diffusion in 2D showing the spatial evolution of a transport variable for a fixed flow field. This model completes the transport equation and also introduces the discretization model proposed by Patankar [4], which implements a multi-scheme model for the convective scheme. The model will be validated against the **Smith-Hutton Case** benchmark [6]. This section will also require the implementation of a new BiConjugate Gradient Stabilized (BiCGSTAB) numerical solver algorithm to solve the complexities introduced by the convective term.

3. The **Navier-Stokes** model will evaluate a coupled 2D Navier-Stokes solver. This model will introduce the concepts of Fractional Step Method (FSM) and Staggered Meshes, requiring a severe overhaul for the code structure. The NS equation model can be simplified in multiple ways, depending on the physical phenomena trying to be described. As such, two different test scenarios of increasing complexity will be evaluated. The first one consists of the **Lid Driven Cavity Case** benchmark [7], which simplifies the model to include only the mass conservation component and the 2 momentum conservation components. The second case consists of the **Differentially Heated Cavity Case** benchmark [8], which introduces an additional energy conservation component. Fractional Step Method (FSM) and Staggered Meshes will also be introduced and their effect in model convergence will be studied.

1 Pure Diffusion Solver

This chapter will explain the implementation of a pure diffusion model [5, 2], which is the simplest version of a transport equation introduced in Chapter . This scenario is obtained by setting the velocity field to zero ($\mathbf{v} = 0$), which eliminates the convective term from the expression. This case describes the physical phenomena of heat conduction through solid bodies, where energy is transmitted exclusively through diffusion as described by Fourier's Law, shown in Eq. (8). The simplicity of the model lends itself to be a good introduction for studying the numerical discretization and validating the core infrastructure of the code implementation. The core components **Material**, **Mesh**, **Discretizer**, **Solver**, and **Probe** were defined on this stage and will remain as the fundamental structure of the model moving forward.

$$q_{cond} = -\lambda \nabla T \tag{8}$$

Following the previously mentioned assumption of a stationary flow field, the convective term can be removed from the governing transport equation, reducing it to the expression shown in Eq. 9. For a heat conduction phenomena, the diffusivity Γ corresponds to the thermal conduction coefficient λ, while the source term S_P corresponds to the internal heat generation q_V. As such, the expanded transport equation becomes Fourier's heat equation, shown in Eq. (9).

$$\rho c_P \frac{\partial T}{\partial t} = \nabla(\lambda \nabla T) + q_V \tag{9}$$

Integrating Eq. (9) over the control volume V_P and a time step Δt, and applying the Gauss divergence theorem to convert the diffusion term from a volume integral into a surface integral across the cell faces, gives:

$$\rho c_P V_P \frac{T_P - T_P^0}{\Delta t} = \oint_{\partial V_P} \lambda \nabla T \cdot \hat{n}\, dA + q_V V_P \tag{10}$$

The surface integral is approximated as a discrete sum over the four cell faces. The face-normal gradient at each interface is evaluated by a central difference between the adjacent cell centroids, and the face conductivity λ_k is computed from the harmonic mean of the adjacent cell values (Eq. (13)):

$$\oint_{\partial V_P} \lambda \nabla T \cdot \hat{n}\, dA \approx \lambda_w S_w \frac{T_W - T_P}{d_{PW}} + \lambda_e S_e \frac{T_E - T_P}{d_{PE}} + \lambda_s S_s \frac{T_S - T_P}{d_{PS}} + \lambda_n S_n \frac{T_N - T_P}{d_{PN}} \tag{11}$$

Substituting Eq. (11) into Eq. (10) yields the fully discretized equation for node P:

$$\rho c_P V_P \frac{T_P - T_P^0}{\Delta t} = \lambda_w S_w \frac{T_W - T_P}{d_{PW}} + \lambda_e S_e \frac{T_E - T_P}{d_{PE}} + \lambda_s S_s \frac{T_S - T_P}{d_{PS}} + \lambda_n S_n \frac{T_N - T_P}{d_{PN}} + q_V V_P \tag{12}$$

Coefficient	Expression
a_W	$\frac{\lambda_w S_w}{d_{PW}}$
a_E	$\frac{\lambda_e S_e}{d_{PE}}$
a_S	$\frac{\lambda_s S_s}{d_{PS}}$
a_N	$\frac{\lambda_n S_n}{d_{PN}}$
b_P	$a_P^0 T_P^0 + q_V V_P$

Table 1: Summary of discretized coefficients for standard FVM formulation for an interior cell. Where a_k corresponds to the coefficient describing the interaction with an adjacent node, and b_P is the right-hand side (RHS) term that describes the influence of the previous step and the source.

An important consideration for pure-diffusion problems is the evaluation of the face diffusivity Γ_k when a cell face lies at a material interface. A simple arithmetic mean would overestimate the conductance between two dissimilar materials; the correct choice is the harmonic mean, which is consistent with the series-resistance analogy for heat conduction:

$$\Gamma_k = \frac{2\Gamma_P \Gamma_K}{\Gamma_P + \Gamma_K} \tag{13}$$

For the thermal conduction model of this chapter, $\Gamma \equiv \lambda$.

By expanding Eq. (12) and reorganizing the terms the same expression shown in Eq. (6) can be obtained for this particular problem. Table 1 shows a summary of the discretized coefficient expressions

1.1 Boundary Conditions

Since boundary conditions are imposed instead of corresponding to the interaction of the center node with a neighboring node, their implementation in the code needs a special formulation. The current model supports 3 different types of boundary conditions; namely **Dirichlet**, **Neumann**, and **Robin**. The explanation concerning the implementation of boundary conditions will be done taking a west boundary as reference; the same concept detailed in this scenario will apply for the other boundaries with their respective coefficients.

Dirichlet boundary conditions directly assign the variable value at the boundary. As such, the gradient can be approximated by using the distance between the cell centroid and the boundary wall d_{Pw} and following the same discretization definitions shown in Table 1. The coefficient term will be added to the diagonal coefficient a_P and the assigned temperature is imposed by introducing it as part of b_P.

$$a_W = 0 \qquad a_P \mathrel{+}= D_w = \frac{\lambda_w S_w}{d_{Pw}} \qquad b_P \mathrel{+}= D_w T_{BC} \tag{14}$$

Neumann boundary conditions assign the flux at the surface rather than the magnitude. As such, the flux value is already known at the surface and can be directly included into the RHS term of the discretization. No additional terms need to be added to the diagonal coefficient.

$$a_W = 0 \qquad b_P \mathrel{+}= q_{BC} S_w \tag{15}$$

Robin boundary conditions, also known as convection / hybrid boundary conditions, impose a variable heat flux that relates the temperature at the node T_P with the temperature of a far-away fluid T_∞ through a convective heat transfer coefficient h. The resulting conductance term is added to the diagonal coefficient a_P and to the RHS of the discretized equation.

$$a_W = 0 \qquad a_P \mathrel{+}= h S_w \qquad b_P \mathrel{+}= h S_w T_\infty \tag{16}$$

Implementation and validation of the 1D and 2D cases are discussed in the following subsections.

1.2 1D Transient Conduction - 1D Rod Case

In this section, the implementation of the heat conduction model for the **1D Rod Case** will be developed. The problem consists of a thin $1m$ bar with adiabatic walls on the south/north faces and Dirichlet boundary conditions for the west/east faces. The west wall is set at $T_w = 100°C$, while the eastern wall is set at $T_e = 20°C$, with initial conditions set at $T_0 = 30°C$. Furthermore, the rod is defined as being made of copper with properties $\lambda = 400W/(m \cdot K)$, $\rho = 8960kg/m^3$, and $c_P = 380J/(kg \cdot K)$. A schematic of the problem is shown in Figure 2.

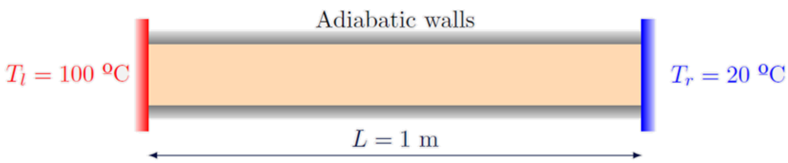

Figure 2: Diagram of 1D Rod Case, including original boundary conditions proposed for that case [9].

1.2.1 Model Implementation

The model implementation was designed to work using a configuration file that includes all the relevant parameters required to define a simulation. This ensures that the model will be robust enough to handle a wide range of simulations without needing any modifications. A summary of all the different parameters included in the file is shown in Table 2. A detailed explanation of each one of the groups and their relevant information is shown below. **Physical** includes the initial temperature and a list of the available materials and their thermophysical properties. **Geometrical** includes the geometric parameters; the model can only support rectangular shapes in this implementation and changes will be made when more complex geometries are studied. Sections must be defined with a material, the number of nodes for that specific section N_s, beginning and ending coordinates x_0 and x_1, and internal heat generation q_V. **Numerical** includes the numerical parameters; the model currently has both the Gauss-Seidel (GS) and Conjugate Gradient (CG) solvers implemented and both will be compared as a first step before performing the relevant studies discussed ahead. **Mesh** includes the option to choose from four different algorithms, as well as all relevant parameters for each of these alternatives. **Boundary Conditions** includes an array describing the boundaries for the model and can currently support Dirichlet, Neumann, and Convection boundary conditions. In addition, it supports variable temperatures for Dirichlet boundary conditions.

Group	Parameter	Description
Physical Data	T0	Initial value for temperature map.
	materials	List of vectors with thermophysical properties corresponding to each material.
Geometrical Data	width	Geometric dimension.
	height	Geometric dimension.
	sections	List of vectors indicating the distribution of materials in the simulation. Also includes amount of nodes and internal heat generation.
Numerical Data	scheme	Scheme for simulation.
	solver	Solver algorithm for simulation.
	maxIterations	Limit of iterations per timestep.
	endTime	Last instant of the simulation.
	timeStep	Time increment after each instant.
	tolNumeric	Tolerance for numerical solver.
	tolTemporal	Tolerance for steady-state convergence.
Mesh Data	meshAlgorithm	Algorithm for mesh generation.
	strength	Cubic algebraic mesh parameter.
	centering	Cubic algebraic mesh parameter.
	kappa	Power law mesh parameter.
	delta	Hyperbolic tangent mesh parameter.
Boundary Data	boundaries	List of vectors describing the boundary conditions on each side.

Table 2: Configuration file structure for 1D Rod Case. Showing the design structure for the metadata used to define the simulations.

1.2.2 Temporal Scheme

One of the first numerical parameters to be evaluated in this project is the temporal scheme. This parameter determines the relevance that each time-step will have in the calculation of the next one. The **explicit** scheme, relies exclusively on information available from the current time-step, described as a function $f(T^{n-1})$ and requires a small enough time-step to ensure stability, given by Eq. (17). On the other hand, the **implicit** scheme relies exclusively on information from the time-step being currently calculated, being a function of $f(T^n)$. Finally, the **Crank-Nicolson** scheme employs a second-order scheme that averages the estimation made by both the explicit and implicit temporal schemes, being a function of $f(0.5(T^{n-1}+T^n))$.

$$\Delta t_{max} = \frac{\Delta x^2}{2\alpha} \tag{17}$$

The convergence residual per time step for all three schemes is compared in Fig. 3. Crank-Nicolson achieves the lowest residual at every time step while remaining stable beyond the explicit stability limit, making it the preferred temporal scheme for all subsequent diffusion studies.

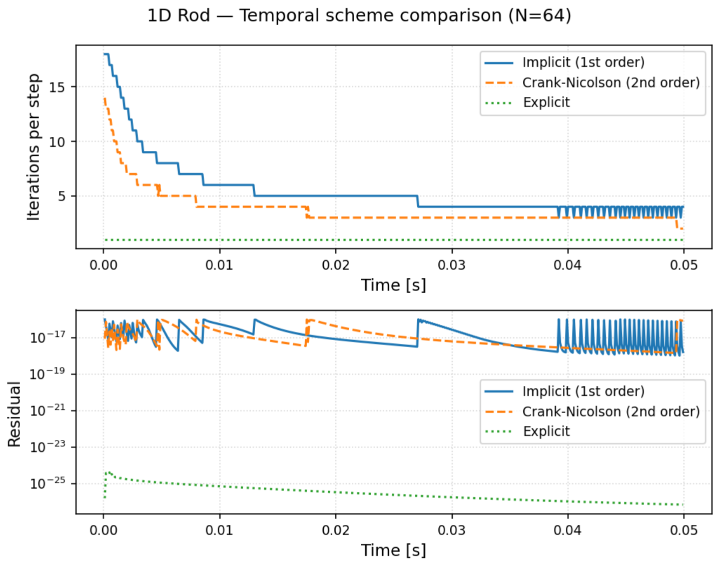

Figure 3: Comparison of explicit, implicit, and Crank-Nicolson temporal schemes ($N = 64$, $\Delta t = 1 \times 10^{-4}$ s, CG solver). Top: inner iterations required to reach convergence at each time step. Bottom: convergence residual $\|\mathbf{b} - A\mathbf{T}\|$. The explicit scheme requires no linear solve (zero iterations); implicit and CN carry the same per-step cost but allow arbitrarily large Δt.

1.2.3 Solver Comparison

An additional decision to make when solving this kind of problems is the numerical solver to be used during simulations. While the matrix definition remains constant in both cases, as both require a sparse matrix in the form $A\mathbf{T} = b$, selecting the right solver can have a big impact in the performance of the solver. In this project, an initial comparison between the Gauss-Seidel (GS) and Conjugate Gradient (CG) algorithms will be made with the objective of evaluating performance and efficiency. A brief introduction to both algorithms is shown below before showing the results of said comparison.

Gauss-Seidel The GS method solves linear systems by iteratively cycling through each node in sequence and updating them by using the most recent values from its adjacent nodes. Each cell needs to be calculated directly by using Eq. (18), which delays the propagation of information from one cell to the others. While convergence is a guarantee when the matrix is diagonally dominant, it is susceptible to mesh size as it will directly impact the number of operations.

$$T_P^{n+1} = \frac{1}{a_P}\left(b_P - \sum_{k \neq P} a_k T_k^*\right) \tag{18}$$

Conjugate Gradient The CG method is an iterative Krylov subspace solver designed specifically for Symmetric Positive Definite (SPD) systems. For heat conduction problems, this matches the diffusion coefficient matrix since the interaction between two adjacent nodes is identical in

Algorithm 1 Gauss-Seidel iterative solver

Require: Coefficient matrix A, RHS vector $\mathbf{b}$, initial guess $\mathbf{T}^0$, tolerance ε, maximum iterations $k_{\max}$
Ensure: Solution vector $\mathbf{T}$

```
T ← T^0
for k = 1 to k_max do
    for i = 1 to N do
        T_i ← (1/a_ii) (b_i − Σ_{j≠i} a_ij T_j)
    end for
    if ||b − AT|| < ε then
        break
    end if
end for
return T
```

$$T_i \leftarrow \frac{1}{a_{ii}}\left(b_i - \sum_{j\neq i} a_{ij}\,T_j\right)$$

both directions (symmetrical) and is diagonally dominant due to the accumulation term in a_P (positive definite). Contrary to the GS method, rather than updating the values one by one, CG minimizes the residual over the sequence of search directions $\mathbf{p}_k$ that are mutually A-conjugate:

$$\mathbf{p}_i^T A \mathbf{p}_j = 0, \qquad i \neq j \tag{19}$$

This guarantees that no iteration undoes the progress made by previous ones. Because of this capacity of propagating information, CG converges in maximum N iterations, being N the amount of unknowns. It is also important to highlight that CG is only valid for SPD systems. As it will be explained in Chapter 2, once symmetry is broken due to the convective term, the solver algorithm needs to be changed to a more generalized variant to be able to handle non-SPD scenarios.

Algorithm 2 Conjugate Gradient solver

Require: SPD coefficient matrix A, RHS vector $\mathbf{b}$, initial guess $\mathbf{T}^0$, tolerance ε, maximum iterations $k_{\max}$
Ensure: Solution vector $\mathbf{T}$

$\mathbf{r}^0 \leftarrow \mathbf{b} - A\mathbf{T}^0;\quad \mathbf{p}^0 \leftarrow \mathbf{r}^0;\quad \mathbf{T} \leftarrow \mathbf{T}^0$
for $k = 0, 1, 2, \ldots, k_{\max}$ **do**
  $\alpha_k \leftarrow \dfrac{\mathbf{r}^k \cdot \mathbf{r}^k}{\mathbf{p}^k \cdot A\,\mathbf{p}^k}$
  $\mathbf{T}^{k+1} \leftarrow \mathbf{T}^k + \alpha_k\,\mathbf{p}^k$
  $\mathbf{r}^{k+1} \leftarrow \mathbf{r}^k - \alpha_k\,A\,\mathbf{p}^k$
  if $\|\mathbf{r}^{k+1}\| < \varepsilon$ **then**
    break
  end if
  $\beta_k \leftarrow \dfrac{\mathbf{r}^{k+1} \cdot \mathbf{r}^{k+1}}{\mathbf{r}^k \cdot \mathbf{r}^k}$
  $\mathbf{p}^{k+1} \leftarrow \mathbf{r}^{k+1} + \beta_k\,\mathbf{p}^k$
end for
return $\mathbf{T}^{k+1}$

Results Solver performance is evaluated using two metrics: the number of inner iterations required to reach convergence at each time step, and the final residual $\|\mathbf{b} - A\mathbf{T}\|$ after convergence. The comparison is carried out on the **1D Rod Case** [9] with Dirichlet boundary conditions ($T_L = 100$, $T_R = 0$), internal heat generation $q_V = 200\,\mathrm{W\,m^{-3}}$, $N = 64$ nodes, $\Delta t = 0.01\,\mathrm{s}$, an implicit temporal scheme, and a uniform mesh. The results are shown in Figure 4.

CG outperforms GS by a considerable margin on both metrics: at the first time step CG converges in 63 iterations (residual $\approx 10^{-31}$, near machine precision), while GS requires 832 iterations to barely satisfy the same tolerance. Averaged over the full simulation, CG needs 21.1 iterations per step versus 669 for GS — approximately 32 times fewer. This is consistent with the structural properties described above: the diffusion matrix is SPD, so CG propagates information globally through its A-conjugate search directions and is guaranteed to converge in at most N steps. GS updates nodes one at a time and therefore requires many more sweeps to resolve long-range temperature gradients. Based on these results, CG was selected as the solver for all subsequent numerical studies in this chapter.

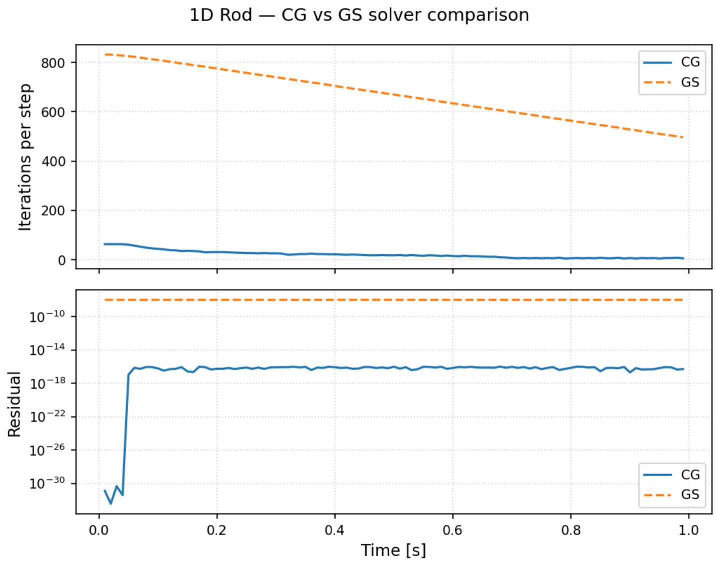

Figure 4: Performance comparison between Gauss-Seidel (GS) and Conjugate Gradient (CG) solvers on the 1D rod case ($N = 64$, $\Delta t = 0.01\,\mathrm{s}$, implicit, $q_V = 200\,\mathrm{W\,m^{-3}}$). Top: number of inner iterations per time step. Bottom: solver residual $\|\mathbf{b} - A\mathbf{T}\|$ at convergence. CG converges in $\approx 32\times$ fewer iterations than GS due to the SPD structure of the diffusion matrix.

1.2.4 Numerical Parameters Study

Two formal convergence studies are carried out to quantify the accuracy of the 1D solver. The temporal convergence study fixes $N = 128$ and varies Δt to isolate the first-order temporal error of the implicit scheme and the second-order temporal accuracy of Crank-Nicolson. The spatial convergence study fixes $\Delta t = 5 \times 10^{-6}\,\mathrm{s}$ (small enough to make the temporal error negligible) and varies N to measure the L_2 error against the transient analytical solution at $t^* = 0.1\,\mathrm{s}$. The parameter ranges are summarised in Table 3.

Temporal Convergence (Δt Sweep) Figure 6 shows the L_2 error against the transient analytical solution as a function of Δt at $N = 128$, with $q_V = 200\,\mathrm{W\,m^{-3}}$. The implicit scheme follows the expected first-order behaviour: error grows from 1.28×10^{-2} at $\Delta t = 10^{-4}\,\mathrm{s}$ to 1.17

Parameter	Values
Δt [s] (temporal)	10^{-4}, 5×10^{-4}, 10^{-3}, 5×10^{-3}, 10^{-2}
N (spatial)	8, 16, 32, 64, **128**

Table 3: Parameter ranges for the convergence studies. Temporal study: Δt swept at fixed $N = 128$ (implicit and CN). Spatial study: N swept at fixed $\Delta t = 5\times10^{-6}$ s (implicit). Bold indicates the mesh used for the temporal study.

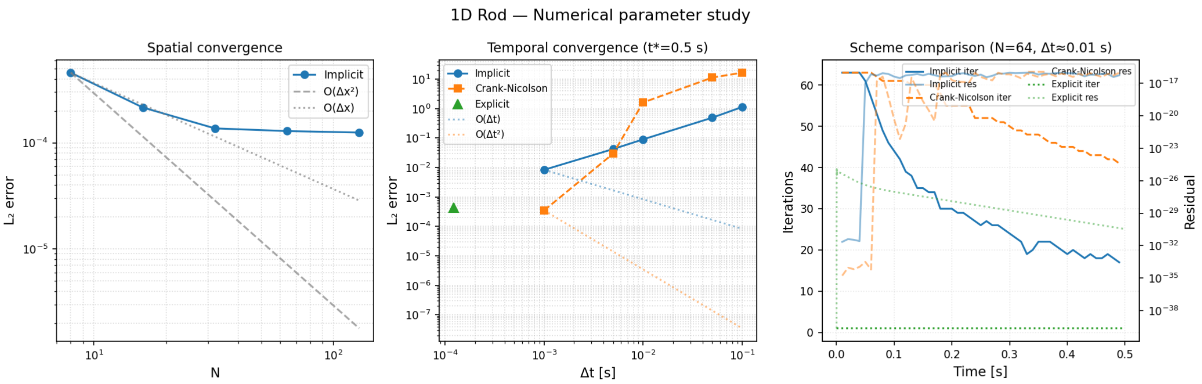

Figure 5: Overview of the 1D numerical parameter study (1D rod, CG solver). Inner CG iterations per time step and convergence residual as functions of time step Δt (left) and mesh size N (right). Crank-Nicolson consistently outperforms the implicit scheme on both metrics at the same parameter values.

at $\Delta t = 10^{-2}$ s, with apparent rates of 0.96–1.00. The Crank-Nicolson scheme achieves second-order temporal accuracy; at $N = 128$ the spatial discretisation error ($L_2 \approx 1.05\times10^{-3}$) sets a floor that the CN temporal error falls below for all stable time steps — the CN curve is therefore flat, indicating that temporal accuracy is not the bottleneck at this spatial resolution. This demonstrates the fundamental advantage of CN: even its largest stable Δt keeps the temporal error well below the spatial error, so the scheme effectively reaches the spatial accuracy limit for free.

Spatial Convergence (N Sweep) Figure 7 shows the L_2 error against the analytical solution as a function of N on a log-log scale, with $\Delta t = 5\times10^{-6}$ s chosen to make the temporal error negligible and $q_V = 200\,\mathrm{W\,m^{-3}}$. The FV scheme achieves the expected second-order spatial convergence: apparent rates of 2.00, 1.97, and 1.88 are measured at $N = 16$, 32, and 64 respectively. At $N = 128$ the rate drops to 1.59 because the temporal error ($O(\Delta t)$ with $\Delta t = 5\times10^{-6}$ s) begins competing with the small spatial error at the finest mesh. The results confirm that the FVM delivers the formal $O(\Delta x^2)$ spatial accuracy expected from the central-differencing of the diffusion flux.

1.2.5 Model Validation

In addition to the Dirichlet boundary condition established in the problem statement, Neumann and Convection boundary conditions have also been implemented in the model. Furthermore, support for variable temperatures for Dirichlet boundary conditions was implemented. To validate this implementation, a comparison will be made with the analytical solution for a steady-state scenario with the transient-state model once it reaches equilibrium.

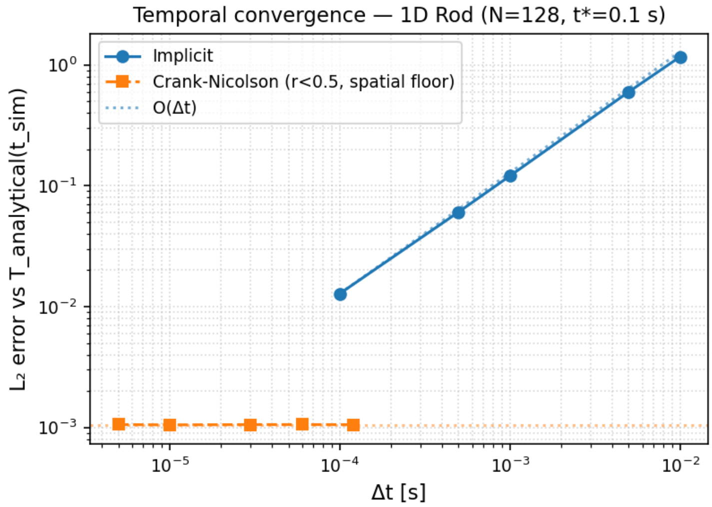

Figure 6: Temporal convergence study (1D rod, $N = 128$, CG solver, $q_V = 200\,\mathrm{W\,m^{-3}}$). L_2 error vs. transient analytical solution at $t^* = 0.1$ s as a function of Δt on a log-log scale. Implicit: first-order convergence ($\sim O(\Delta t)$). Crank-Nicolson: flat curve at the spatial accuracy floor ($L_2 \approx 1.05 \times 10^{-3}$), indicating that the CN temporal error is negligible for all stable Δt values at this mesh.

At steady state the time derivative vanishes and the 1D governing equation reduces to:

$$\frac{d}{dx}\left(\lambda \frac{dT}{dx}\right) + q_V = 0 \tag{20}$$

For uniform λ and $q_V = 0$, this integrates to a linear temperature profile $T(x) = A + Bx$, with constants A and B determined by the two boundary conditions. The test performed in this project will instead add internal heat generation, which modifies the analytical solution to Eq. (21). Table 4 summarises the closed-form solutions for each of the four boundary-condition pairs tested, showing the coefficient values for each case.

$$C_2 + C_1 x - 0.5\frac{q_V V x^2}{\lambda} \tag{21}$$

A summary of the pairs of boundary conditions used for the tests is shown in Table 5, while comparisons with the steady-state solution are shown in Figure 8. As shown, all four pairs accurately reproduce the analytical parabolic profiles. An additional Case 5 (Table 5) is included to validate the time-varying Dirichlet capability: the configuration file accepts a mathematical expression $T(t)$ in place of a constant value, enabling time-dependent thermal boundary conditions without code changes.

1.3 2D Transient Conduction - 4 Materials Case

In this section, the implementation of a heat conduction model for the **4 Materials Case** will be developed. The problem consists of the rectangular domain shown in Fig. 9, which is composed

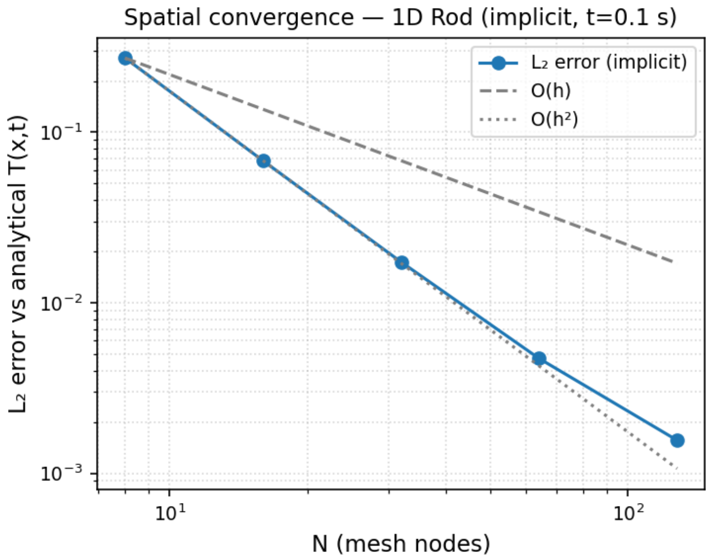

Figure 7: Spatial convergence study (1D rod, $\Delta t = 5 \times 10^{-6}$ s, implicit, CG solver). L_2 error vs. analytical solution at $t^* = 0.1$ s as a function of N on a log-log scale. Reference lines for $O(\Delta x)$ and $O(\Delta x^2)$ are shown. The scheme achieves second-order convergence up to $N = 64$; the slight reduction at $N = 128$ reflects the temporal error floor becoming significant.

by 4 different materials with their thermophysical properties being listed in Table 6. Furthermore, the domain has an initial value of $T_0 = 8°C$ and is bound by the boundary conditions shown in Table 7.

1.3.1 Model Implementation

The model developed for the 2D case will follow the same discretization proposed earlier for the 1D case. However, some modifications to the code structure were needed to support the new functionalities. For example, Probe was expanded to support different types of probes with a better control over the region of the domain that is desired. A summary of the modified parameters included in this iteration of the model is shown in Table 8, summarizing the new parameters in the configuration file. As shown, Probes have now been added as a configurable parameter to allow the user to choose which data is being saved. In addition, refinement was modified to be defined independently for each axis and in separate regions to allow the user better control over the mesh. Furthermore, a Medic class was also included to handle diagnostics and validation of energy conservation in the system. This class is also used as the central tool for debugging as it can easily be modified to log any information relevant to the model.

1.3.2 Numerical Parameter Study

Two studies are carried out to assess the impact of mesh resolution and temporal scheme on solver performance for the 4 Materials Case. The first sweeps N at fixed $\Delta t = 50$ s to quantify how CG iteration count and residual scale with mesh size (Panel A of Fig. 10). The second compares the three temporal schemes (implicit, Crank-Nicolson, and explicit) at fixed $N = 64$ (Panel B). The explicit scheme requires the time step to satisfy the 2D stability limit of Eq. (22), which is much stricter than the $\Delta t = 50$ s used by the other schemes. The mesh sizes tested are summarised in Table 9.

Case	Boundary Conditions	Analytical Solution $T(x)$
1	Neumann (W) – Dirichlet (E)	$C_1 = -\frac{q_{BC}}{\lambda}$ $C_2 = T_{Dir} + 0.5 * \frac{q_V V}{\lambda} - C_1$
2	Robin (W) – Dirichlet (E)	$C_2 = \frac{T_{Dir} + 0.5\frac{q_V V}{\lambda} + \frac{hT_\infty}{\lambda}}{1+h}$ $C_1 = -h\frac{T_\infty - C_2}{\lambda}$
3	Dirichlet (W) – Neumann (E)	$C_2 = T_{Dir}$ $C_1 = \frac{q_V V - q_{BC}}{\lambda}$
4	Dirichlet (W) – Robin (E)	$C_2 = T_{Dir}$ $C_1 = \frac{\frac{hT_\infty}{\lambda} + (1 + 0.5\frac{hq_V V}{\lambda^2}) - \frac{hC_2}{\lambda}}{1 + \frac{h}{\lambda}}$

Table 4: Analytical steady-state solutions for each boundary-condition pair. $\dot{q}_W$ and $\dot{q}_E$ denote the prescribed heat flux entering the domain at the west and east faces respectively; h is the convective heat transfer coefficient and T_∞ the far-field fluid temperature.

$$\Delta t_{max} = \frac{\Delta x^2 \Delta y^2}{2\alpha(\Delta x^2 + \Delta y^2)} \tag{22}$$

N Study (Panel A) Panel A of Fig. 10 shows CG iteration count and convergence residual as a function of N at fixed $\Delta t = 50$ s (implicit). As the mesh is refined from $N = 16$ to $N = 128$, the iteration count grows roughly as $O(\sqrt{N})$, consistent with the spectral properties of CG on 1D-structured systems. The residual increases slightly with N as the larger linear system contains more unknowns per solve.

Scheme Comparison (Panel B) Panel B compares implicit, Crank-Nicolson, and explicit schemes at $N = 64$ over the full simulation time of 15000 s. Implicit and CN both use $\Delta t = 50$ s and complete the simulation in approximately 300 time steps; the explicit scheme is stability-limited to $\Delta t \approx \Delta x^2/(2\alpha_{\min})$, requiring roughly $375\,000$ steps to reach $t = 15000$ s. Despite the drastically different step counts, all three schemes converge to the same steady state. For a stiff diffusion problem with time constant $\tau \approx 1685$ s, the implicit and CN schemes are dramatically more efficient per unit of simulated time.

1.3.3 Model Validation

As a preliminary step, the three boundary condition types (Dirichlet, Neumann, and Robin) were independently verified in 2D by applying an adiabatic Neumann condition ($\partial T/\partial n = 0$)

Case	Boundary Conditions
1	Neumann-Dirichlet
2	Convection-Dirichlet
3	Dirichlet-Neumann
4	Dirichlet - Convection
5	$T_{(t)} = 100 + 50\cos\left(\frac{0.01t}{5\pi}\right)$

Table 5: *Cases for model validation. Each case describes a Left-Right pair of boundary conditions to be tested with the 1D Rod Case.*

Material	ρ $[kg/m^3]$	c_P $[J/(kg \cdot K)]$	λ $[W/(mK)]$
M1	1500	750	160
M2	1600	770	140
M3	1900	810	200
M4	2500	930	140

Table 6: Thermophysical properties of the four materials in the domain.

on the boundaries of one axis, effectively reducing the problem to a 1D configuration along the other axis. The results successfully replicated the steady-state profiles validated in Section 1.2 for each axis, confirming that the boundary condition implementation is correct in two dimensions.

The primary benchmark for this chapter is the **4 Materials Case** [10], whose results are shown in Fig. 11. Since this is a transient simulation, validation is performed by comparing the numerical temperature profile at $t = 15000\,\mathrm{s}$ against the analytical piecewise-linear steady-state solution $T_{\mathrm{ss}}(x)$, which can be derived from the one-dimensional heat-flux continuity condition across the four material interfaces. The overall L_2 error is 5.70×10^{-5}, and the interface temperatures agree to within $0.13\,\mathrm{K}$ of the analytical values ($T(0.25) = 76.55$ vs. $76.63\,°\mathrm{C}$; $T(0.50) = 48.38$ vs. $48.25\,°\mathrm{C}$; $T(0.75) = 28.25$ vs. $28.38\,°\mathrm{C}$).

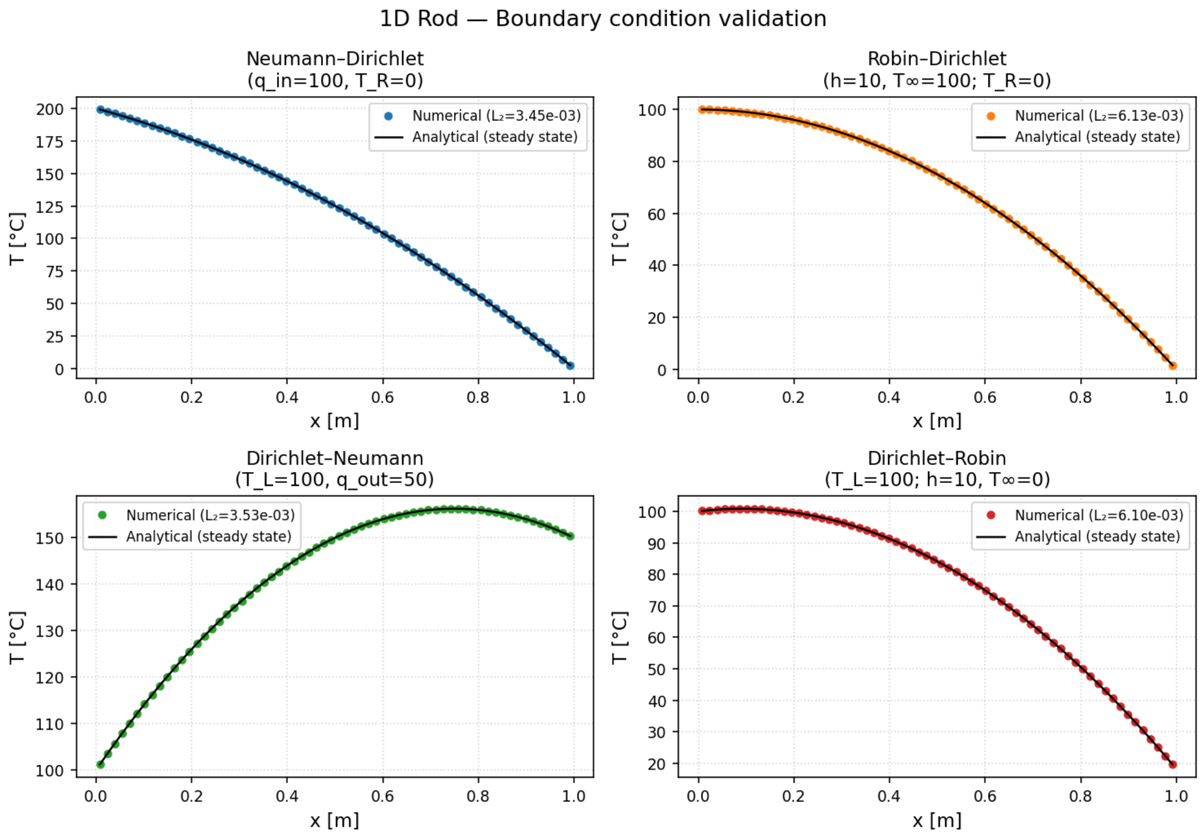

Figure 8: Computed temperature profiles at $t = 4\,\mathrm{s}$ against analytical steady-state solutions for Cases 1–4 (2×2 grid: Neumann–Dirichlet, Robin–Dirichlet, Dirichlet–Neumann, Dirichlet–Robin), all with $q_V = 200\,\mathrm{W\,m^{-3}}$. All profiles are parabolic and reproduce the analytical solution accurately. Neumann cases reach $L_2 \approx 3.5 \times 10^{-3}$; Robin cases give $L_2 \approx 6 \times 10^{-3}$ because at $q_V = 200\,\mathrm{W\,m^{-3}}$ the convective flux at the Robin boundary vanishes identically, reducing the effective BC to a zero-flux Neumann condition and extending the relaxation time constant to $\tau_1 = 4/\pi^2 \approx 0.41\,\mathrm{s}$.

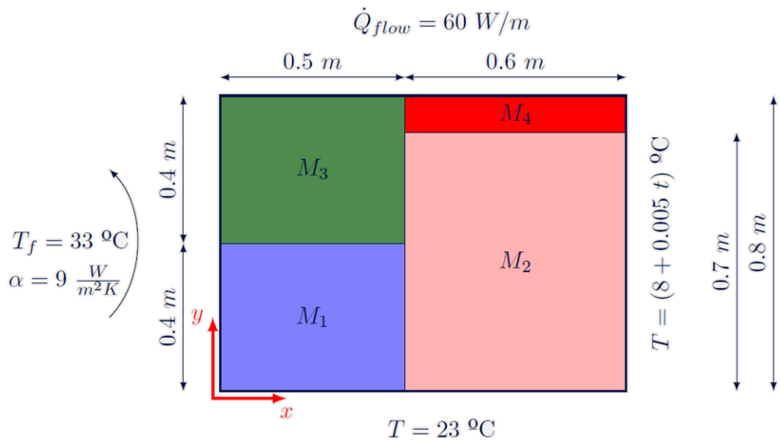

Figure 9: Diagram for 4 Materials Case.

Cavity Wall	Boundary Condition
West	**Robin**: In contact with fluid at $T_f = 33°C$ and $h = 9\,W/(m^2K)$
East	**Dirichlet**: Uniform temperature $T(t) = 8 + 0.005t°C$
South	**Dirichlet**: Uniform temperature $T = 23°C$
North	**Neumann**: Uniform $\dot{Q}_{flow}$

Table 7: Boundary conditions for the 4 Materials Case domain.

Group	Parameter	Description
Mesh Data	N	List specifying total number of nodes for each axis.
	refinement	List of vectors specifying refinement configuration for each axis.
Probe Data	probes	List of vectors specifying the probes configured for the simulation.

Table 8: Configuration file structure changes, highlighting the expansion in mesh refinement capacity and probe flexibility.

Parameter	Values
N	16, 32, **64**, 128

Table 9: Mesh sizes tested in the 4 Materials spatial refinement study (implicit, $\Delta t = 50\,\mathrm{s}$). Bold indicates the resolution used for the benchmark and scheme comparison.

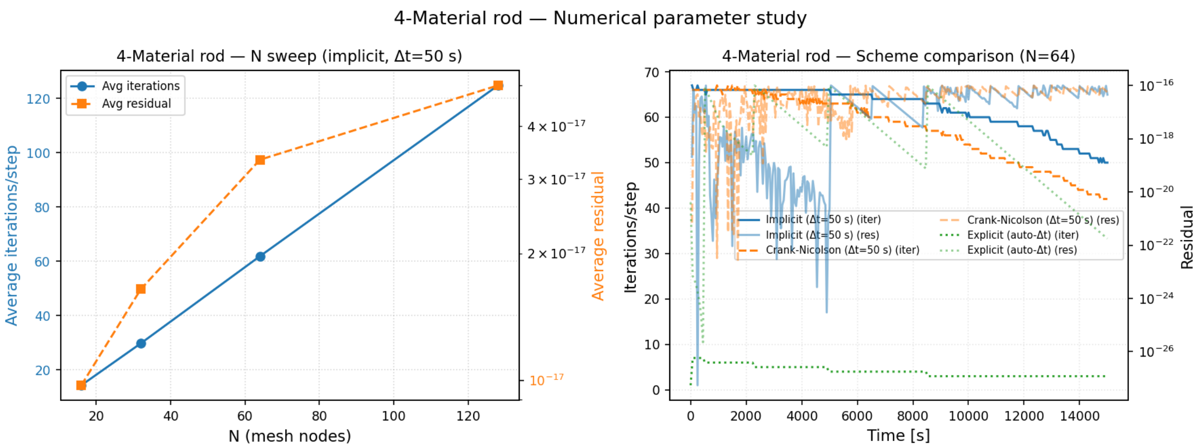

Figure 10: Numerical parameter study for the 4 Materials Case. Panel A (left): average CG iterations per time step and average residual as a function of mesh size N (implicit, $\Delta t = 50\,\mathrm{s}$). Panel B (right): iteration count and residual vs. time for the three temporal schemes at $N = 64$; the explicit scheme requires $\sim 10^3\times$ more time steps than implicit/CN to reach the same simulated time.

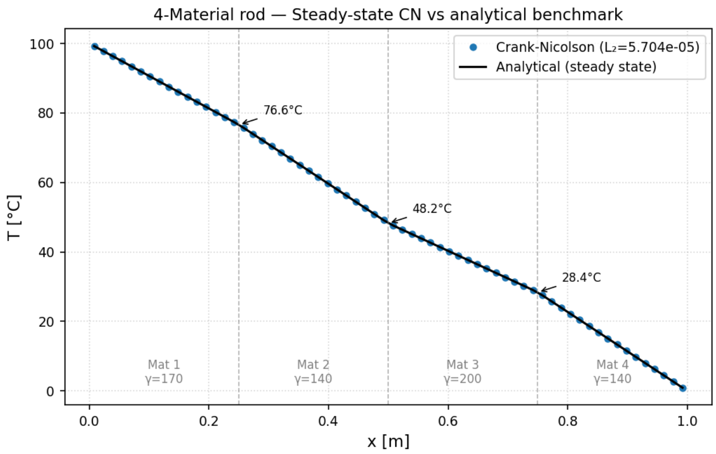

Figure 11: Steady-state temperature profile at $t = 15000\,\mathrm{s}$ (CN, $N = 64$, $\Delta t = 10\,\mathrm{s}$) compared against the analytical piecewise-linear solution $T_{\mathrm{ss}}(x)$ of the 4 Materials Case [10]. Vertical dashed lines indicate material interfaces; slope changes at the interfaces reflect the different thermal conductivities.

2 Convection-Diffusion Model

This chapter extends the pure-diffusion solver of Chapter 1 to the full convection-diffusion transport equation by activating the convective term. While the diffusive term is elliptic and distributes information in all directions, the convective term is hyperbolic: it propagates information strictly in the direction of the flow. This directional nature introduces new numerical challenges. A central-difference approximation of the convective flux is second-order accurate but produces oscillations when convection dominates diffusion; a pure upwind approximation is stable at any Peclet number but reduces to first-order accuracy. Patankar [4] proposed a generalised formulation that unifies these and intermediate schemes through a single coefficient template, allowing the analyst to select the scheme most appropriate for the problem at hand. The model will be validated against the Smith-Hutton benchmark [6], a widely used test for convection-diffusion discretisations.

Activating the convective term in the general transport equation (Eq. (1)) gives the convection-diffusion governing equation:

$$\frac{\partial(\rho\phi)}{\partial t} + \nabla \cdot (\rho \mathbf{v} \phi) = \nabla \cdot (\Gamma \nabla \phi) + S_\phi \tag{23}$$

where ϕ is the transported scalar, $\mathbf{v}$ is the prescribed velocity field, and Γ is the diffusivity. Unlike Chapter 1 where $\phi \equiv T$ and $\Gamma \equiv \lambda$, both quantities are kept general here since the solver is first validated against a passive scalar transport benchmark before being extended to the Navier-Stokes model in Chapter 3.

Integrating Eq. (23) over the control volume V_P and a time step Δt, and applying the divergence theorem to both the convective and diffusive terms — following the same procedure as in Chapter 1 — converts them into face integrals. At each face k, the result is a diffusive contribution $D_k(\phi_K - \phi_P)$ and a convective contribution $F_k\phi_k$. The two face quantities that characterize these contributions are defined as follows.

The **diffusive conductance** D_k measures the ease of diffusive transport across face k:

$$D_k = \frac{\Gamma_k S_k}{d_{PK}} \tag{24}$$

The **convective flux** F_k carries the mass flow rate through face k, defined positive in the direction of the outward normal:

$$F_k = \rho\, v_k\, S_k \tag{25}$$

Their ratio defines the cell-face **Peclet number** Pe_k, which quantifies the relative strength of convection over diffusion at each face:

$$Pe_k = \frac{F_k}{D_k} \tag{26}$$

When $|Pe_k| \ll 1$ the transport is diffusion-dominated and the face value ϕ_k is well approximated by a central average of the two adjacent cell values; when $|Pe_k| \gg 1$ the transport is convection-dominated and ϕ_k must be taken from the upwind node to maintain stability. Patankar's formulation encodes this behaviour through the scheme function $A(|Pe|)$, which modulates the diffusive weight so that the combined face flux always satisfies diagonal dominance. Based on this formulation, the discretized coefficients for a 2D interior cell are:

$$a_W = D_w A(|Pe_w|) + \max(F_w,\, 0) \tag{27}$$

$$a_E = D_e A(|Pe_e|) + \max(-F_e,\, 0) \tag{28}$$

$$a_S = D_s A(|Pe_s|) + \max(F_s,\, 0) \tag{29}$$

$$a_N = D_n A(|Pe_n|) + \max(-F_n,\, 0) \tag{30}$$

$$a_P = a_W + a_E + a_S + a_N + a_P^0 - (F_e - F_w + F_n - F_s) \tag{31}$$

$$b_P = S_\phi V_P + a_P^0 \phi_P^0 \tag{32}$$

$$a_P^0 = \frac{\rho V_P}{\Delta t}$$

The $\max(F_k, 0)$ terms ensure that the convective contribution to a_W is active only when flow arrives from the west (i.e. $F_w > 0$), and similarly for the other faces: each neighbour coefficient captures the information carried into cell P by the flow. The last grouped term in a_P is the discrete continuity residual $(F_e - F_w) + (F_n - F_s)$, which represents the net mass flux leaving the cell; for a divergence-free velocity field it vanishes exactly, otherwise it acts as a correction to maintain conservation. A summary of the $A(|Pe|)$ expressions for the implemented schemes is shown in Table 10.

Each entry in Table 10 corresponds to a different face interpolation rule. The **CDS** scheme computes ϕ_k as a linear average of the two adjacent cell values; this is second-order accurate but $A(|Pe|) = 1 - 0.5|Pe|$ becomes negative for $|Pe| > 2$, which violates diagonal dominance and leads to unbounded oscillations. The **UDS** scheme always takes ϕ_k from the upwind cell, giving $A(|Pe|) = 1$ for all Peclet numbers and unconditional diagonal dominance at the cost of first-order accuracy; the truncation error manifests as artificial numerical diffusion proportional to $\rho v_k \Delta x/2$. The **Hybrid** scheme applies CDS where $|Pe_k| \leq 2$ and switches to UDS above that

Scheme	$A(\lvert Pe \rvert)$
CDS	$1 - 0.5\,\lvert Pe \rvert$
UDS	1
Hybrid	$\max\big(0,\ 1 - 0.5\,\lvert Pe \rvert\big)$
Power Law	$\max\big(0,\ (1 - 0.1\,\lvert Pe \rvert)^5\big)$
Exponential	$\dfrac{\lvert Pe \rvert}{e^{\lvert Pe \rvert} - 1}$

Table 10: Scheme function $A(|Pe|)$ for each implemented convective discretisation [4].

threshold — exactly where the CDS coefficient would turn negative — providing a pragmatic stability guarantee with improved accuracy in low-Peclet regions. The **Power Law** scheme transitions more smoothly between regimes through a fifth-order polynomial approximation of the exponential solution. The **Exponential** scheme is the exact analytical solution of the steady 1D convection-diffusion equation at a face, making it the most accurate interpolation available; it is included as a reference baseline but is rarely used in practice due to the cost of evaluating exponentials at every face each iteration.

2.1 Boundary Conditions

The boundary condition framework introduced in Chapter 1 is extended here to account for the convective term. Dirichlet and Neumann conditions are both supported, though the Robin boundary condition is not implemented in this chapter: the physical scenarios studied from this point onwards involve solid walls or prescribed velocity inflows and outflows, none of which require the convective heat transfer coupling that Robin conditions model. A second structural change is the introduction of **ghost nodes**: instead of adding an additional layer of nodes located directly at the wall (as in Chpater 1), the interior nodes located at a half-cell distance from the boundary wall. The boundary condition is imposed by prescribing the value of these ghost nodes, which simplifies the handling of the convective flux since the face lies exactly halfway between the ghost node and the first interior node. The wall distance used in the diffusive conductance D_k is therefore the full spacing between the ghost node and node P. The explanation is given for a west boundary; it applies to all other faces with their respective coefficients.

Dirichlet boundary conditions prescribe the value of ϕ at the wall. Since ϕ_{BC} is known, both the diffusive and convective fluxes through the west face become explicit source terms. For an inflow face ($F_w > 0$, flow entering from the west), the upwind value used by the convective scheme is ϕ_{BC} itself:

$$a_W = 0 \qquad a_P \mathrel{+}= D_w A(|Pe_w|) + \max(F_w,\,0) \qquad b_P \mathrel{+}= \big[D_w A(|Pe_w|) + \max(F_w,\,0)\big]\,\phi_{BC} \tag{33}$$

For an outflow face ($F_w < 0$), the $\max(F_w, 0)$ term vanishes and only the diffusive conductance contributes, consistent with the fact that the flow carries information away from the boundary.

Neumann boundary conditions prescribe the diffusive flux normal to the wall. The most common case in this chapter is the **zero-gradient** condition $\partial\phi/\partial n = 0$, applied at outflow boundaries. This is implemented by setting the ghost node value equal to that of the first interior node, which makes the diffusive conductance contribution vanish identically. The convective flux at the outflow face is then evaluated using the interior node value ϕ_P, which is already implicit in a_P and requires no explicit source addition:

$$a_W = 0 \qquad b_P \mathrel{+}= \dot{q}_{BC}\, S_w \tag{34}$$

For the adiabatic case ($\dot{q}_{BC} = 0$) used at outflow boundaries, the equation simplifies to $a_W = 0$ with no additions to a_P or b_P.

2.2 2D Convection-Diffusion - Smith-Hutton Case

The Smith-Hutton problem [6, 11] is a widely used benchmark for convection-diffusion schemes. It consists of a rectangular domain $[-1, 1] \times [0, 1]$ subject to a prescribed, analytically divergence-free velocity field:

$$u = 2y(1 - x^2), \qquad v = -2x(1 - y^2) \tag{35}$$

This velocity field generates a circular flow that enters along the bottom from the left, sweeps across the domain the domain and exits through the bottom from the right. Because $\nabla \cdot \mathbf{v} = 0$ is satisfied analytically, the discrete continuity correction term in Eq. (31) vanishes exactly, making this a clean test of the convective discretisation alone. A schematic of the problem is shown in Fig. 12.

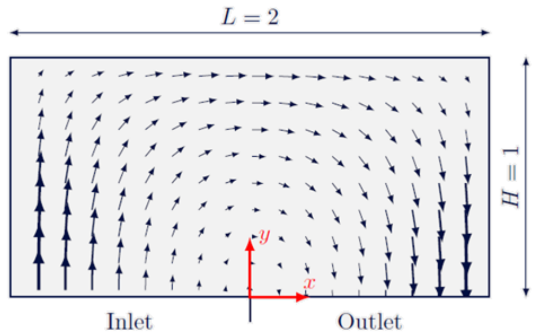

Figure 12: Smith-Hutton problem domain and velocity streamlines [6].

The transport variable ϕ represents a passive scalar carried by the flow. Boundary conditions are:

- **South inlet** ($y = 0,\ -1 \le x \le 0$): Dirichlet profile $\phi = 1 + \tanh\big(10(2x + 1)\big)$, which ramps from 0 at $x = -1$ to 2 at $x = 0$.
- **South outlet** ($y = 0,\ 0 \le x \le 1$): Neumann zero-gradient $\partial\phi/\partial n = 0$.
- **All remaining walls**: Dirichlet $\phi = 0$.

The steep inlet profile makes this a demanding benchmark: a high-order scheme must resolve the sharp gradient without introducing oscillations, while an upwind scheme will smear it due to numerical diffusion. The problem is parameterised by the ratio ρ/Γ, which controls the global

Peclet number. Three cases are tested: $\rho/\Gamma = 10^1$, 10^3, and 10^6, with $\Gamma = 1$ held fixed throughout. Validation is performed by comparing the computed exit profile $\phi(x,\, y = 0)$ for $0 \leq x \leq 1$ against the reference values provided by Smith and Hutton [6].

2.2.1 Model Implementation

The convection-diffusion solver is built on the same modular architecture as Chapter 1, extending it with three key additions. First, the velocity field is no longer a fixed parameter but a prescribed analytical input: the two components u and v are specified as mathematical expressions in the configuration file, which are parsed and evaluated at each cell face during the assembly of the convective fluxes F_k. This approach keeps the model general, allowing any analytically defined, divergence-free velocity field to be tested without modifying the source code. Second, ghost nodes are used at all domain boundaries (see Section 2) instead of the half-cell wall treatment of Chapter 1, enabling a consistent treatment of both the diffusive and convective face fluxes. Third, a new Boundary probe type was added to the Probe class, which records ϕ along a specified boundary face at each saved time-step; this is required for the Smith-Hutton validation, which compares the exit profile at the south outlet.

A summary of the new configuration parameters for this model is shown in Table 11. Compared to the previous chapter, the main additions are the **spatScheme** parameter for the convective scheme, the **velocityField** entry for the analytical velocity expressions, and the **medicOn** flag that activates the diagnostic tool.

Group	Parameter	Description
Numerical Data	spatScheme	Convective scheme selection for simulation.
Physical Data	velocityField	Mathematical expressions for each velocity component.
Medic Data	medicOn	Diagnostic tool control parameter.

Table 11: Configuration file structure for the convection-diffusion model, highlighting the new parameters introduced in this chapter.

2.2.2 BiCGSTAB Solver

The introduction of the convective term breaks the symmetry of the coefficient matrix A: the upwind contribution to a_W is not equal to the corresponding entry in a_E, so $A \neq A^T$ in general. The Conjugate Gradient method requires a symmetric positive-definite system and therefore is no longer applicable once the convective term is present. The BiConjugate Gradient Stabilized (BiCGSTAB) method [12] extends the Krylov subspace approach to general non-symmetric systems by maintaining two coupled residual sequences. The method introduces a shadow residual $\hat{\mathbf{r}}$, fixed at the initial residual $\mathbf{r}^0$ and never updated, which provides a stable projection direction throughout the iteration. Each step of the algorithm contains two substeps: a standard BiCG step that computes a step length α_k, and a local minimisation step controlled by ω_k that minimises the residual along the direction $A\mathbf{s}^k$. This suppresses the irregular convergence oscillations that characterise standard BiCG and gives the method its stabilised name. Algorithm 3 summarises the procedure.

Like CG, BiCGSTAB is a Krylov method and converges in at most N iterations for an $N \times N$ system, though in practice convergence is reached much earlier. The cost per iteration is higher than CG — two matrix-vector products per step instead of one — but this is outweighed by

Algorithm 3 BiConjugate Gradient Stabilized (BiCGSTAB) solver

Require: Non-singular matrix A, RHS vector $\mathbf{b}$, initial guess $\boldsymbol{\phi}^0$, tolerance ε, maximum iterations $k_{\max}$
Ensure: Solution vector $\boldsymbol{\phi}$

$\mathbf{r}^0 \leftarrow \mathbf{b} - A\boldsymbol{\phi}^0; \quad \hat{\mathbf{r}} \leftarrow \mathbf{r}^0; \quad \mathbf{p}^0 \leftarrow \mathbf{r}^0; \quad \boldsymbol{\phi} \leftarrow \boldsymbol{\phi}^0$
for $k = 0, 1, 2, \ldots, k_{\max}$ **do**
  $\alpha_k \leftarrow \dfrac{\hat{\mathbf{r}} \cdot \mathbf{r}^k}{\hat{\mathbf{r}} \cdot A\mathbf{p}^k}$
  $\mathbf{s}^k \leftarrow \mathbf{r}^k - \alpha_k A\mathbf{p}^k$
  if $\|\mathbf{s}^k\| < \varepsilon$ **then**
    $\boldsymbol{\phi} \leftarrow \boldsymbol{\phi}^k + \alpha_k \mathbf{p}^k; \quad$ **break**
  end if
  $\omega_k \leftarrow \dfrac{(A\mathbf{s}^k) \cdot \mathbf{s}^k}{\|A\mathbf{s}^k\|^2}$
  $\boldsymbol{\phi}^{k+1} \leftarrow \boldsymbol{\phi}^k + \alpha_k \mathbf{p}^k + \omega_k \mathbf{s}^k$
  $\mathbf{r}^{k+1} \leftarrow \mathbf{s}^k - \omega_k A\mathbf{s}^k$
  if $\|\mathbf{r}^{k+1}\| < \varepsilon$ **then**
    break
  end if
  $\beta_k \leftarrow \dfrac{\hat{\mathbf{r}} \cdot \mathbf{r}^{k+1}}{\hat{\mathbf{r}} \cdot \mathbf{r}^k} \cdot \dfrac{\alpha_k}{\omega_k}$
  $\mathbf{p}^{k+1} \leftarrow \mathbf{r}^{k+1} + \beta_k(\mathbf{p}^k - \omega_k A\mathbf{p}^k)$
end for
return $\boldsymbol{\phi}$

the ability to handle the asymmetric systems that arise whenever convection is present. As explained in Chapter 3, the same solver is carried forward to the Navier-Stokes model.

2.2.3 Numerical Parameter Study

Before the full scheme comparison can be run under controlled conditions, a working mesh resolution must be established. The mesh refinement study described below uses the Smith-Hutton problem with the UDS scheme and $\rho/\Gamma = 10^3$ as the base case, sweeping the number of nodes N across four levels. The chosen resolution is then used for all subsequent validation runs. A summary of the tested values is shown in Table 12.

Parameter	Values
N	16, 32, **64**, 128

Table 12: Mesh sizes tested in the spatial refinement study. The bold value indicates the resolution selected for validation.

Mesh Refinement The Smith-Hutton problem is solved at four progressively finer uniform meshes using the Hybrid scheme at the most demanding regime $\rho/\Gamma = 10^6$, where convection fully dominates. Since no closed-form analytical solution exists, the L_2 error of the exit profile $\phi(x, y = 0)$ for $0 \le x \le 1$ is computed against the Smith and Hutton reference [6]:

$$e_N = \|\phi_{\text{num}} - \phi_{\text{ref}}\|_2 = \sqrt{\frac{1}{N_{\text{out}}} \sum_i (\phi_i - \phi_{\text{ref},i})^2} \tag{36}$$

where N_{out} is the number of outlet nodes used in the comparison. The left panel of Fig. 13 shows the evolution of the exit profile as N increases; the right panel plots e_N on a log-log scale against the mesh spacing $h = 2/N_x$.

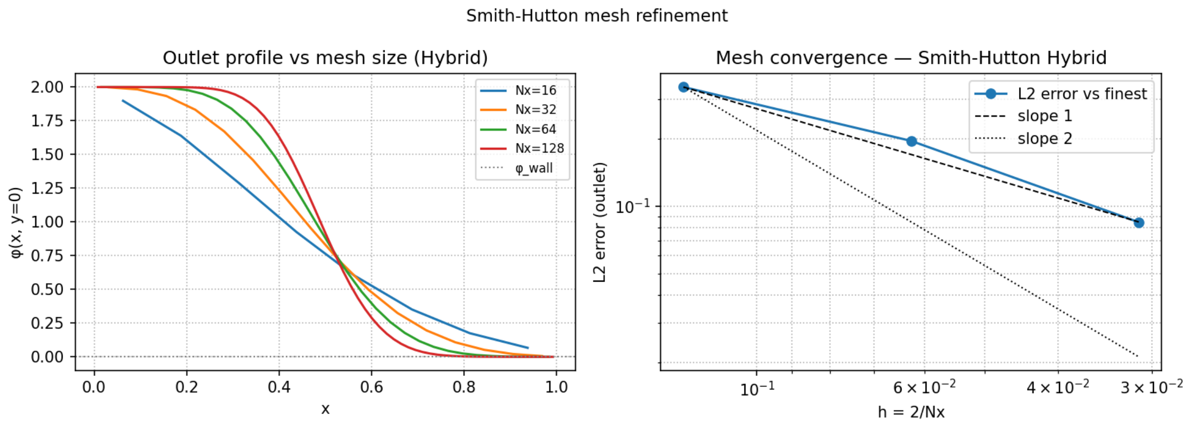

Figure 13: Mesh refinement study for the Smith-Hutton case ($\rho/\Gamma = 10^6$, Hybrid). Left: exit profile $\phi(x, 0)$ for each mesh level, compared against the Smith and Hutton reference. Right: L_2 error against mesh spacing $h = 2/N_x$; reference slope lines indicate first- and second-order convergence rates.

The computed errors decrease from $e_{16} = 0.457$ to $e_{128} = 0.165$, yielding an apparent convergence order $p \approx 0.46$ — close to first order rather than the second-order formal accuracy of Hybrid. The reason is that at $\rho/\Gamma = 10^6$ the local Péclet number $Pe_k \gg 2$ across virtually the entire domain, so the scheme reverts to UDS everywhere and the solution carries first-order numerical diffusion along streamlines. True second-order convergence would require CDS with $Pe_k \leq 2$, which is unachievable at this Péclet regime for any practical mesh. The resolution $N = 64$ is selected for subsequent validation runs as the level at which the exit profile has stabilised in shape. Figure 14 extends the mesh comparison to both UDS and Hybrid; CDS is excluded because it diverges at $\rho/\Gamma = 10^6$ (see Table 12).

2.2.4 Model Validation

The model is validated by running the Smith-Hutton problem for three convective schemes — CDS, UDS, and Hybrid — across all three values of ρ/Γ, using the mesh resolution established by the refinement study. The three schemes represent different points in the accuracy-stability trade-off described in Table 10: CDS is second-order accurate but loses diagonal dominance for $|Pe_k| > 2$; UDS maintains diagonal dominance at all Péclet numbers at the cost of first-order accuracy; and the Hybrid scheme switches between the two at $|Pe_k| = 2$.

Solver performance at the first time step is summarised in Table 13, and the exit profiles $\phi(x, y = 0)$ are compared against the reference values [6] in Figs. 15 and 16. At $\rho/\Gamma = 10$, CDS and Hybrid require identical iteration counts (125) because $Pe_k \ll 2$ everywhere, so Hybrid applies CDS throughout and the two schemes produce the same linear system; UDS differs slightly

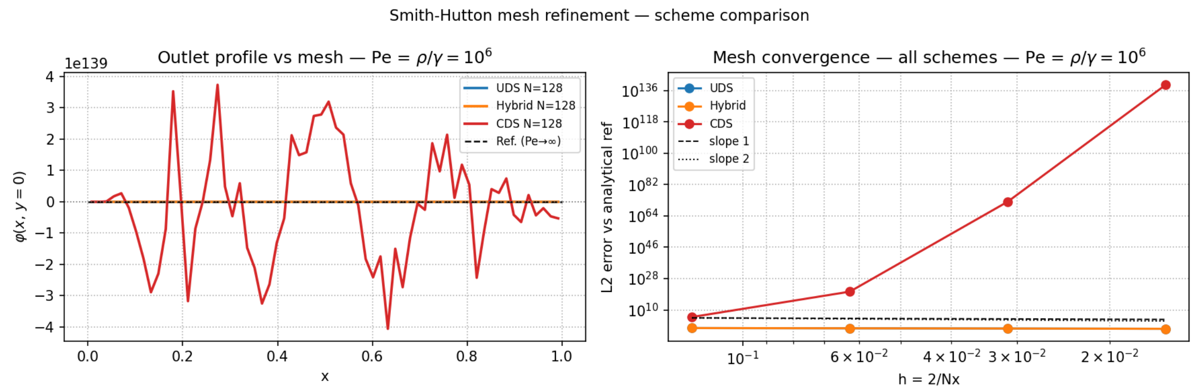

Figure 14: Multi-scheme mesh convergence at $\rho/\Gamma = 10^6$ (UDS and Hybrid; CDS excluded due to instability at $Pe_k \gg 2$). Left: exit profile $\phi(x, 0)$ at $N = 128$ for each scheme, compared against the analytical reference $\phi_{\text{ref}}(x) = 1 + \tanh\big(10(1 - 2x)\big)$. Right: log-log L_2 error vs. mesh spacing h; both schemes converge at approximately first order.

(127 iterations) due to its upwind-biased coefficients. At $\rho/\Gamma = 10^3$, UDS is the fastest to converge (108 iterations) because upwinding reinforces diagonal dominance; Hybrid requires 117 iterations; CDS exceeds the solver iteration limit on the first step as the large off-diagonal convective coefficients make the system ill-conditioned, though the solution recovers by the third time step. At $\rho/\Gamma = 10^6$, CDS diverges numerically; UDS and Hybrid converge in 115 and 122 iterations respectively, with larger absolute residuals reflecting the large magnitude of the convective right-hand side rather than poor convergence — the relative residuals satisfy the 10^{-8} tolerance in both cases.

ρ/Γ	Scheme	Iterations (step 1)	$\lVert\mathbf{r}\rVert$ (step 1)
10^1	CDS	125	1.59×10^{-7}
	UDS	127	6.26×10^{-8}
	Hybrid	125	1.59×10^{-7}
10^3	CDS	$>1999^{\dagger}$	6.59×10^{-4}
	UDS	108	6.56×10^{-7}
	Hybrid	117	9.77×10^{-7}
10^6	CDS	Diverged	
	UDS	115	4.28×10^{-4}
	Hybrid	122	7.24×10^{-4}

Table 13: BiCGSTAB iterations and absolute residual $\lVert\mathbf{r}\rVert$ at the first time step for all three convective schemes across the three ρ/Γ cases ($N = 64$). †CDS at $\rho/\Gamma = 10^3$ hits the solver iteration limit on step 1 due to the ill-conditioned system; the residual shown is after 1999 iterations. The solution recovers by step 3.

At $\rho/\Gamma = 10$ the three schemes produce nearly identical exit profiles: diffusion is strong enough to homogenise any interpolation differences, and all results agree well with the Smith and Hutton reference.

As convection intensifies at $\rho/\Gamma = 10^3$, scheme differences become clearly visible. UDS smears the steep inlet gradient due to first-order numerical diffusion, producing a flatter outlet profile with the peak near $x = 0$ underestimated. Hybrid recovers some sharpness by applying CDS in

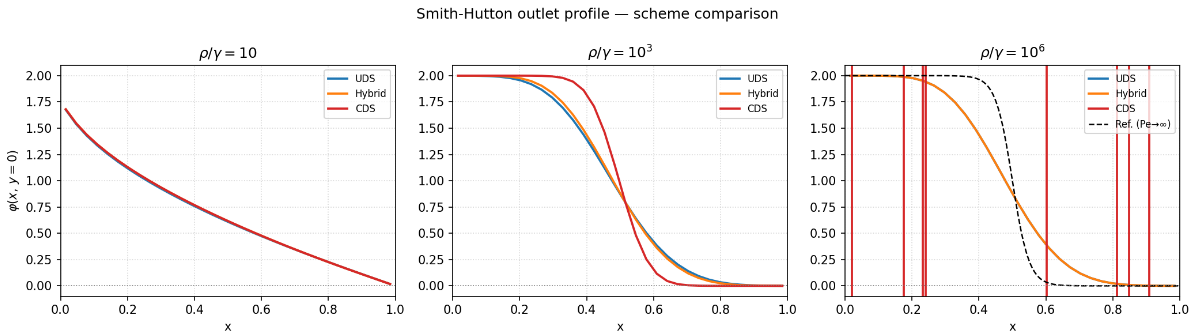

Figure 15: Exit profile $\phi(x, 0)$ for CDS, UDS, and Hybrid compared against the reference of Smith and Hutton [6] at each ρ/Γ case. For $\rho/\Gamma = 10^6$ the reference corresponds to the $Pe \to \infty$ limit.

low-Péclet regions, but reverts to UDS in the high-velocity core, limiting its improvement. CDS, remaining second-order throughout, tracks the reference profile most closely and produces the sharpest outlet gradient.

At $\rho/\Gamma = 10^6$, the local Péclet number far exceeds 2 on every cell face for any practical mesh. The CDS coefficient $A(|Pe|) = 1 - 0.5|Pe|$ becomes strongly negative, breaking diagonal dominance; the solver diverges, as visible in the right column of Fig. 16 where the field is shown at the scale of numerical overflow. Among the stable schemes, UDS produces a near-flat outlet profile that barely reproduces the shape of the analytical reference $\phi_{\mathrm{ref}}(x) = 1 + \tanh(10(1 - 2x))$ for $Pe \to \infty$. Hybrid remains stable by construction and yields a sharper profile than UDS, but still under-captures the steep gradient near $x = 0$ because first-order UDS blending dominates the entire domain at this Péclet level.

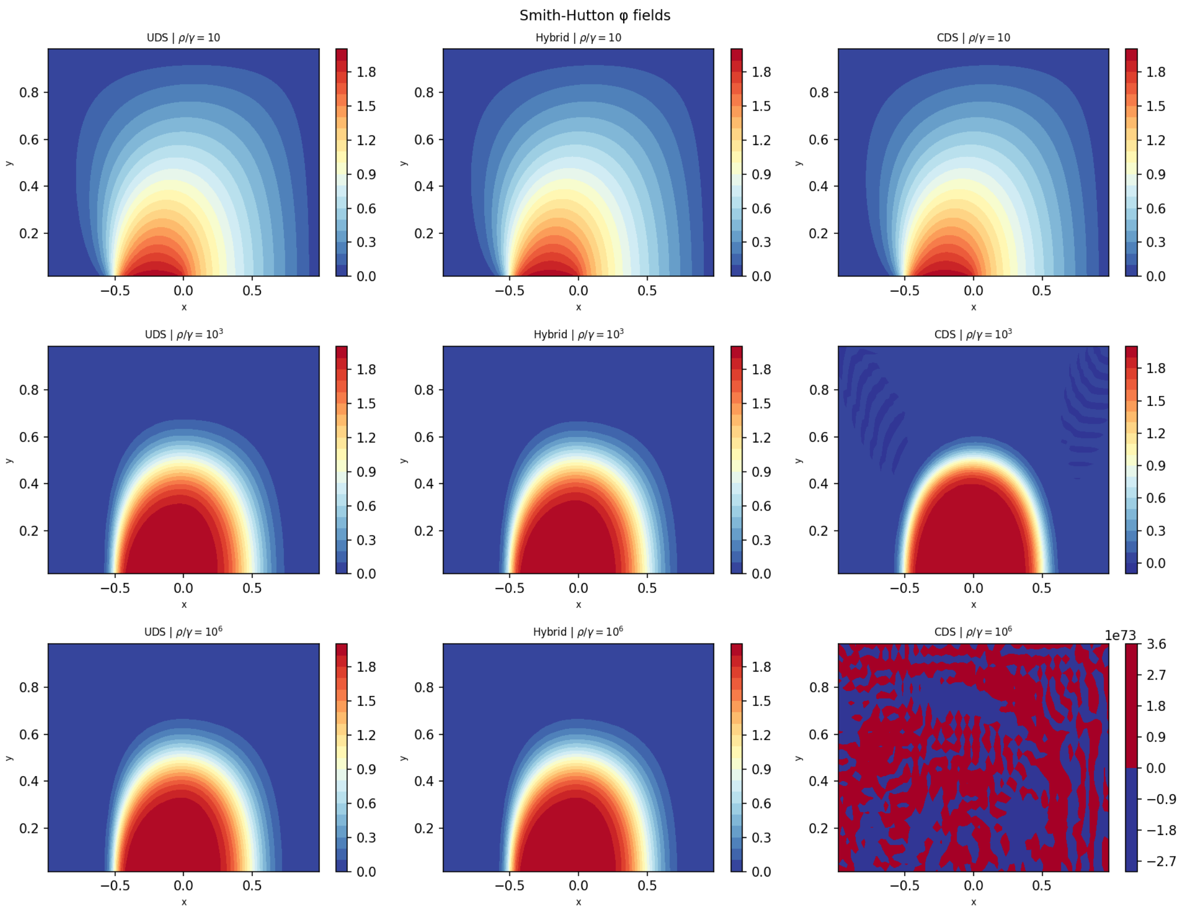

Figure 16: Steady-state ϕ field for UDS (left column), Hybrid (centre column), and CDS (right column) across all three ρ/Γ cases (rows). At $\rho/\Gamma = 10^6$ the CDS solution has diverged and is shown at the scale of the numerical overflow.

3 2D Navier-Stokes Model

The Navier-Stokes equations [1] describe the motion of viscous incompressible fluids and form the foundation of computational fluid dynamics. Unlike the scalar transport problems of Chapters 1 and 2, where a single PDE governs a single unknown, the NS system is a coupled set of equations involving velocity, pressure and temperature.

Two fundamental challenges distinguish this problem from the preceding chapters. The first is **pressure-velocity coupling**: in incompressible flow, pressure does not have its own time-derivative equation. Instead, it acts as a Lagrange multiplier that enforces the continuity constraint $\nabla \cdot \mathbf{v} = 0$, and its value at each time step must be determined implicitly from that constraint rather than marched forward independently. The second is the **nonlinear convective term** $\rho(\mathbf{v}\cdot\nabla)\mathbf{v}$ in the momentum equation, which couples the velocity components to each other and prevents the direct application of the linear solvers developed in the previous chapters.

Both challenges are addressed in combination. The **Adams-Bashforth** scheme evaluates the convective and diffusive terms explicitly using values from the current and previous time steps, so that the right-hand side of the momentum equation is fully known without inner iteration. The **Fractional Step Method** [3] then decouples the pressure from the velocity by splitting each time step into a predictor–Poisson–corrector sequence, reducing the coupled system to a set of independent linear problems of the type already established in the previous chapters.

The governing equations for incompressible viscous flow are:

$$\nabla \cdot \mathbf{v} = 0 \tag{37}$$

$$\rho \frac{\partial \mathbf{v}}{\partial t} + \rho (\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla P + \mu \nabla^2 \mathbf{v} \tag{38}$$

$$\rho c_p \frac{\partial T}{\partial t} + \rho c_p \nabla \cdot (\mathbf{v}\, T) = \nabla \cdot (\lambda \nabla T) + \dot{q}_V \tag{39}$$

Equations (37) and (38) govern the velocity and pressure fields in all test cases. Equation (39) is the energy transport equation, which shares the same structure as the general transport equation of Chapter with $\phi \equiv T$ and $\Gamma \equiv \lambda$; it is only activated for the Differentially Heated Cavity (Section 3.4), where temperature differences drive the flow through buoyancy.

The time discretization of Eq. (38) uses the **Adams-Bashforth** method [3], a second-order explicit multistep scheme. The key idea is to approximate the combined convective-diffusive operator $R(\mathbf{v})$ at the unknown time level $n + 1$ by linear extrapolation from the two preceding known levels n and $n - 1$:

$$R(\mathbf{v}) = \mu \nabla^2 \mathbf{v} - \rho (\mathbf{v} \cdot \nabla)\mathbf{v} \tag{40}$$

$$\rho\,\frac{\mathbf{v}^{n+1}-\mathbf{v}^{n}}{\Delta t} = -\nabla P^{n+1} + \frac{3}{2}\,R(\mathbf{v}^{n}) - \frac{1}{2}\,R(\mathbf{v}^{n-1}) \tag{41}$$

The coefficients $3/2$ and $-1/2$ follow from a second-order backward extrapolation of R using the two known levels; this maintains temporal second-order accuracy while keeping the entire right-hand side (except for ∇P^{n+1}) explicit in known quantities. The spatial discretization of $R(\mathbf{v})$ reuses the framework of the previous chapters: the diffusive term $\mu\,\nabla^2\mathbf{v}$ is discretized as in Chapter 1, and the convective term $\rho\,(\mathbf{v}\cdot\nabla)\mathbf{v}$ uses Patankar's scheme with the selected $A(|Pe|)$ function from Chapter 2. The unknown pressure P^{n+1} is handled by the Fractional Step Method described in the following subsection. Table 14 shows the changes done to the configuration structure for this model, which reformulat the definition of initial conditions and boundary conditions for each variable. In addittion, a new probe type was introduced (**Field**) to target the velocity field directly.

Group	Parameter	Description
Physical Data	T0	Initial value for temperature map.
	P0	Initial value for pressure map.
	VF0	List with initial value for each velocity component.
Boundary Data	boundariesEnergy	Boundary conditions for temperature solver.
	boundariesPressure	Boundary conditions for pressure solver.
	boundariesVelocity	Boundary conditions for each velocity component.

Table 14: Configuration file structure for the Navier-Stokes model. The main additions with respect to Chapter 2 are the separate initial conditions and boundary conditions for velocity and pressure.

3.1 Fractional Step Method

The Fractional Step Method resolves the pressure-velocity coupling in Eq. (41) by decomposing each time step into three sequential operations. The core observation is that if the pressure gradient is temporarily omitted from the momentum update, the resulting *predictor* velocity $\mathbf{v}^*$ will not in general satisfy the continuity equation; the mismatch between $\mathbf{v}^*$ and the correct $\mathbf{v}^{n+1}$ is then used to derive a scalar Poisson equation for P^{n+1}, whose solution closes the system.

Step 1 — Predictor. The velocity is advanced from $\mathbf{v}^n$ to the intermediate field $\mathbf{v}^*$ using the Adams-Bashforth evaluation of R, with the pressure gradient omitted:

$$\mathbf{v}^* = \mathbf{v}^n + \frac{\Delta t}{\rho}\left[\frac{3}{2}\,R(\mathbf{v}^n) - \frac{1}{2}\,R(\mathbf{v}^{n-1})\right] \tag{42}$$

The field $\mathbf{v}^*$ is fully determined from known quantities and is computed explicitly without solving a linear system. At the discrete level, the x-momentum equation for each staggered u-node P is written in terms of the FVM residual $\mathcal{R}_u^n$ evaluated using Patankar's unified scheme (Table 10):

$$u_P^* = u_P^n + \frac{\Delta t}{\rho\, V_P}\left[\frac{3}{2}\,\mathcal{R}_u^n - \frac{1}{2}\,\mathcal{R}_u^{n-1}\right] \tag{43}$$

$$\mathcal{R}_u^n = \sum_{k\in\{w,e,s,n\}} a_k^n\,(u_k^n - u_P^n), \qquad a_k^n = D_k\,A(|Pe_k^n|) + \max(-F_k^n,\,0) \tag{44}$$

where the face diffusivities, convective fluxes, and face Péclet numbers are:

$$D_k = \frac{\mu\, S_k}{d_{Pk}}, \qquad F_k^n = \rho\,\bar{v}_k^n\, S_k, \qquad Pe_k^n = \frac{F_k^n}{D_k} \tag{45}$$

The face velocity $\bar{v}_k^n$ driving each convective flux is not stored at the faces of the u-control volume; it is obtained by linear interpolation between the two surrounding staggered nodes, reducing to arithmetic averaging on a uniform mesh. The y-momentum equation for each staggered v-node follows the same structure with v replacing u and the roles of the x- and y-directions exchanged.

Step 2 — Pressure Poisson equation. Subtracting Eq. (42) from the full update Eq. (41) gives the velocity correction:

$$\mathbf{v}^{n+1} = \mathbf{v}^* - \frac{\Delta t}{\rho}\,\nabla P^{n+1} \tag{46}$$

Taking the divergence of Eq. (46) and imposing the incompressibility constraint $\nabla\cdot\mathbf{v}^{n+1} = 0$ yields the pressure Poisson equation:

$$\nabla^2 P^{n+1} = \frac{\rho}{\Delta t}\,\nabla\cdot\mathbf{v}^* \tag{47}$$

Equation (47) is a standard diffusion-type equation with the divergence of the predictor as its source term. Integrating over V_P and applying the divergence theorem — following the same procedure as in Chapter 1 — gives:

$$a_P P_P = a_W P_W + a_E P_E + a_S P_S + a_N P_N + b_P \tag{48}$$

with coefficients $a_k = S_k/d_{Pk}$ for $k \in \{w, e, s, n\}$ (equivalent to setting $\Gamma = 1$ in Table 1), $a_P = \sum_k a_k$, and:

$$b_P = -\frac{\rho}{\Delta t}\left(u_e^*\,S_e - u_w^*\,S_w + v_n^*\,S_n - v_s^*\,S_s\right) \tag{49}$$

This is a symmetric positive-definite linear system (same structure as the diffusion equation) and is solved by BiCGSTAB, consistent with Chapter 2.

Step 3 — Velocity corrector. With P^{n+1} known from Step 2, the corrected velocity field is obtained from Eq. (46). On the staggered mesh, each velocity node sits at a face between two adjacent pressure cells, so the discrete pressure gradient links immediately neighbouring pressure nodes without interpolation. For a u-node at the east face between pressure cells P and E, and a v-node at the north face between cells P and N:

$$u_P^{n+1} = u_P^* - \frac{\Delta t}{\rho} \frac{P_E^{n+1} - P_P^{n+1}}{\Delta x} \tag{50}$$

$$v_P^{n+1} = v_P^* - \frac{\Delta t}{\rho} \frac{P_N^{n+1} - P_P^{n+1}}{\Delta y} \tag{51}$$

Both updates are explicit once P^{n+1} is known and require no linear solve. By construction, $\mathbf{v}^{n+1}$ satisfies the continuity equation to the accuracy of the Poisson solve.

Energy equation. Equation (39) is discretized following the same FVM procedure as the general transport equation derived in the Introduction (Eqs. (6)–(5)), with $\phi \equiv T$ and $\Gamma \equiv \lambda$. The coefficient expressions are identical in form to those of Chapter 2 and are not reproduced here.

3.1.1 Staggered Meshes

Discretizing pressure and velocity on the same collocated grid leads to a well-known pathology. The central-difference approximation of the pressure gradient at cell P in the x-direction is:

$$\left.\frac{\partial P}{\partial x}\right|_P \approx \frac{P_E - P_W}{2\Delta x} \tag{52}$$

This formula samples pressure nodes two cells apart and ignores the immediate neighbours. As a consequence, a *checkerboard* pressure field (one that alternates sign between adjacent cells) produces a zero gradient at every node. The momentum equation cannot distinguish this spurious oscillatory pattern from a physically uniform pressure field, and the solver cannot suppress it.

The staggered mesh arrangement [3] eliminates this decoupling by relocating the velocity components to the cell faces. Pressure nodes remain at cell centres; horizontal velocity u nodes are placed at the east/west face midpoints; and vertical velocity v nodes are placed at the north/south face midpoints, as shown in Fig. 17. With this layout, the pressure gradient driving the u-node at the east face of cell P is:

$$\left.\frac{\partial P}{\partial x}\right|_e \approx \frac{P_E - P_P}{\Delta x} \tag{53}$$

which links adjacent pressure nodes directly. Similarly, the discrete divergence at cell P:

$$\nabla \cdot \mathbf{v}\big|_P \approx \frac{u_e - u_w}{\Delta x} + \frac{v_n - v_s}{\Delta y} \tag{54}$$

is computed from face velocities that are the immediate neighbours of the cell centroid. Both odd-even decouplings are eliminated without any interpolation between mesh levels.

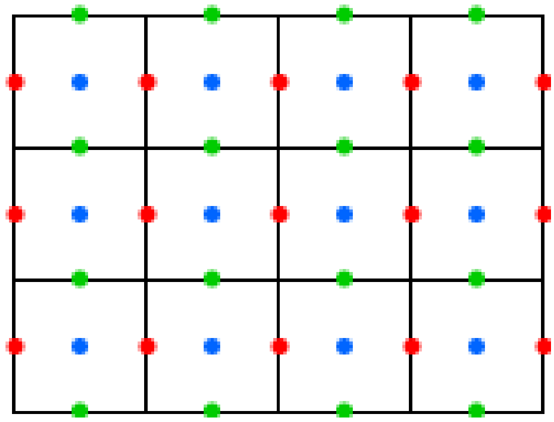

Figure 17: Staggered mesh layout. Pressure nodes (blue) are located at cell centres; u-velocity nodes (red) at east/west face midpoints; v-velocity nodes (green) at north/south face midpoints.

3.2 Boundary Conditions

Boundary conditions for the Navier-Stokes model must be specified separately for the velocity and for the pressure, since the two fields are governed by different equations within the FSM algorithm.

Velocity boundary conditions. All solid walls impose the **no-slip** condition, which sets both velocity components to zero at the wall:

$$u = 0, \qquad v = 0 \qquad \text{(solid wall)} \tag{55}$$

For the Lid-Driven Cavity, which will be explained further below, the moving top wall prescribes a non-zero tangential velocity while maintaining zero normal velocity:

$$u = U, \qquad v = 0 \qquad \text{(moving lid)} \tag{56}$$

Both conditions are Dirichlet constraints and are imposed through the same coefficient modification used in Chapter 2: the known wall velocity enters a_P and b_P of the corresponding momentum equation cell. On the staggered mesh, the velocity node lies exactly at the face midpoint, so no ghost-node distance correction is required for the wall-tangent component; the normal-to-wall component is zero by the no-penetration condition and is enforced directly.

Pressure boundary conditions. The pressure Poisson equation (Eq. (47)) has no prescribed Dirichlet value at any wall: every physical boundary is solid, so $\mathbf{v}^{n+1} \cdot \hat{n} = 0$ on all wall faces. Differentiating Eq. (46) in the wall-normal direction and applying the no-penetration condition yields the natural boundary condition for pressure:

$$\frac{\partial P^{n+1}}{\partial n} = 0 \qquad \text{(all walls)} \tag{57}$$

A purely Neumann Poisson problem has a one-dimensional null space: P^{n+1} is determined only up to an additive constant. To obtain a unique solution, the pressure is fixed to zero at a single reference node (a domain corner), which does not affect the velocity field since only ∇P enters the corrector step.

3.3 Lid-Driven Cavity

The Lid-Driven Cavity is a classical benchmark for incompressible viscous flow solvers [7, 13]. The problem consists of a square domain $\Omega = [0, L] \times [0, L]$ with $L = 1$ (dimensionless) in which the top wall moves at constant velocity $U = 1$ in the x-direction while all other walls are stationary. No thermal effects are present; the flow is driven entirely by the shear stress imparted by the moving lid, which sets up a large primary recirculation vortex that fills the cavity. A schematic of the problem is shown in Fig. 18.

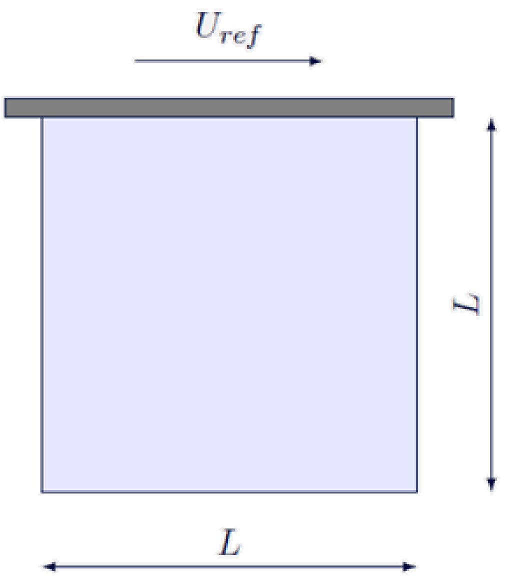

Figure 18: Lid-Driven Cavity schematic. The top wall moves at velocity U while the remaining three walls are stationary.

The flow regime is characterised by the Reynolds number:

$$Re = \frac{\rho\, U\, L}{\mu} \tag{58}$$

As Re increases, inertial effects shift the primary vortex centre downstream and secondary counter-rotating vortices develop in the cavity corners. The reference case is $Re = 1000$, which lies in the inertia-dominated regime and presents sufficiently high local Peclet numbers to expose differences between convective schemes. The reference solution is provided by Ghia et al. [7], who tabulate the u-velocity along the vertical mid-plane $x = 0.5$ and the v-velocity along the horizontal mid-plane $y = 0.5$.

3.3.1 Model Implementation

The Lid-Driven Cavity uses the full NS solver described above with the energy equation (Eq. (39)) deactivated, since the problem is isothermal. The domain is a single uniform material with $\rho = 1$ and viscosity $\mu = 1/Re$, so that $L = U = 1$ satisfies Eq. (58) by construction. Boundary conditions follow Section 3.2: no-slip on the west, east, and south walls; moving lid ($u = 1,\ v = 0$) on the north wall. Initial conditions are $\mathbf{v}^0 = \mathbf{0}$ and $P^0 = 0$ throughout the domain. An additional probe type "Field" was added for the model, which record the full 2D distribution of u and v at each saved time step.

3.3.2 Numerical Parameter Study

Two independent parameter studies are carried out before the validation runs. The first establishes the minimum uniform mesh resolution that captures the primary vortex structure. The second assesses whether a non-uniform mesh, clustered toward the walls using the hyperbolic tangent distribution controlled by the δ parameter, provides accuracy gains over a uniform mesh at the same node count.

Mesh Refinement Four uniform meshes are swept for both the CDS and Hybrid schemes at $Re = 1000$. The minimum u-velocity along the vertical centreline $x = 0.5$, denoted $u_{\min}$, is tracked as a scalar convergence metric and compared against the Ghia reference [7]. The tested mesh sizes are summarised in Table 15, and the computed $u_{\min}$ values for both schemes are listed in Table 16.

Parameter	Values
N	16, 32, **64**, 128

Table 15: Mesh sizes tested in the spatial refinement study for the Lid-Driven Cavity ($Re = 1000$). Bold value indicates the resolution selected for subsequent studies.

N	CDS		Hybrid	
	$u_{\min}$	Error	$u_{\min}$	Error
16	-0.349	–	-0.158	–
32	-0.337	–	-0.246	–
64	-0.361	5.70%	-0.309	19.24%
128	-0.367	4.13%	-0.354	7.55%
Extrap. (∞)	-0.369	3.60%	-0.461	–
Ghia	-0.38289			

Table 16: Minimum u-velocity $u_{\min}$ at $x = 0.5$ for CDS and Hybrid schemes at four mesh levels ($Re = 1000$). Error is computed against the Ghia reference. Richardson extrapolation gives apparent orders $p = 2.00$ (CDS) and $p = 0.50$ (Hybrid).

The CDS results, shown in Fig. 19, exhibit second-order convergence ($p = 2.00$), consistent with the formal accuracy of the scheme. The grid convergence index (GCI) at $N = 128$ is 0.68%, confirming that the finest-mesh solution is well-resolved. The extrapolated limit $u_{\min}^{\infty} = -0.369$ lies 3.60% below the Ghia reference of -0.38289. This residual offset is structural: Ghia et al. used a stream-function/vorticity formulation that eliminates pressure entirely, whereas the FSM pressure-correction step introduces a small systematic perturbation to the velocity field that does not vanish with mesh refinement.

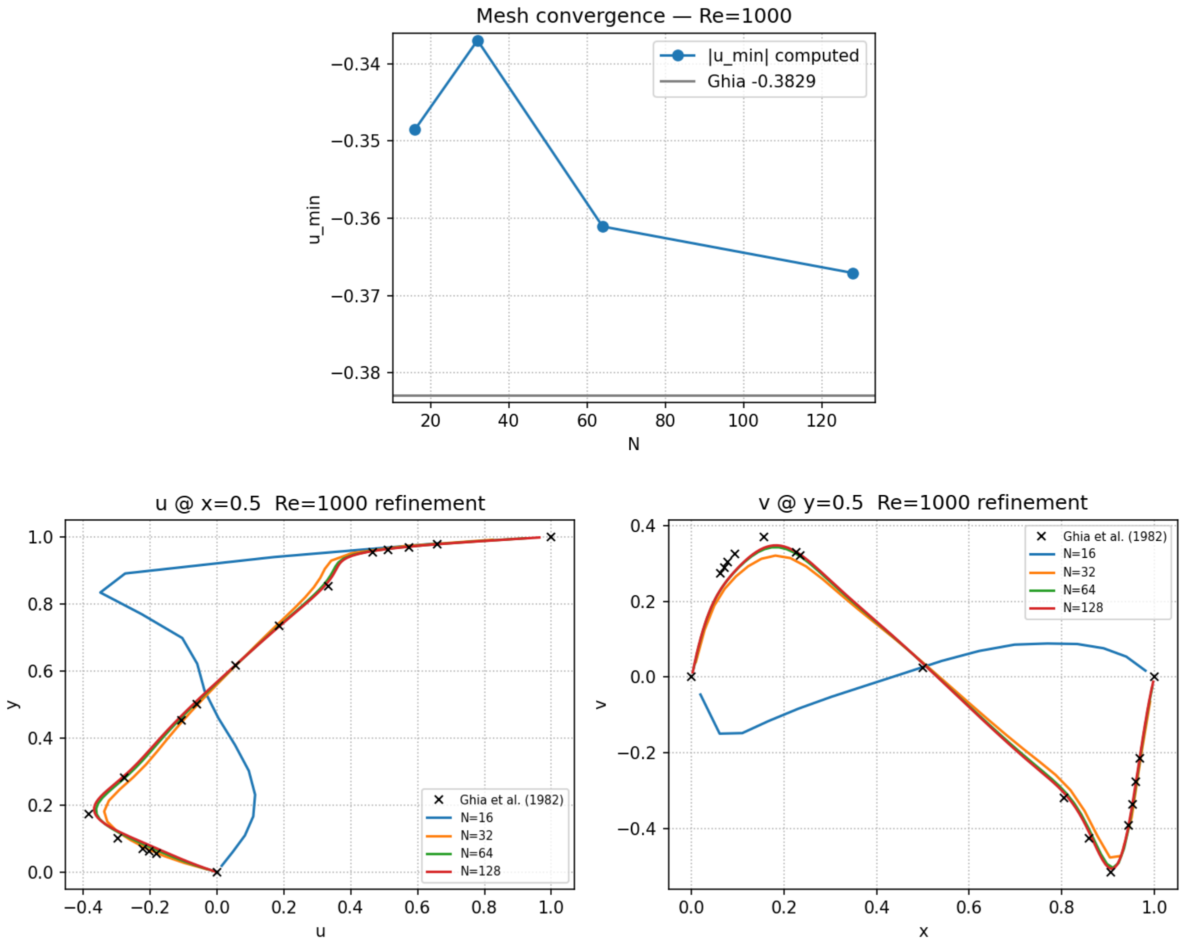

Figure 19: Mesh refinement study for the Lid-Driven Cavity ($Re = 1000$, CDS). Top: minimum u-velocity $u_{\min}$ at $x = 0.5$ as a function of N, compared against the Ghia reference. Bottom: centreline u (left) and v (right) profiles at each mesh level.

The Hybrid scheme converges considerably more slowly: Richardson extrapolation yields $p \approx 0.50$ and a GCI of 37.79% at $N = 128$, indicating that the finest Hybrid solution is far from grid-independent. The slow convergence reflects the UDS blending that activates wherever $Pe_k > 2$: in the high-velocity lid region and near the primary vortex, first-order numerical diffusion smears the velocity gradients and systematically underestimates $|u_{\min}|$. The corresponding profiles are shown in Fig. 20.

Non-uniform Mesh Study With the uniform resolution established, the effect of near-wall clustering is assessed. The hyperbolic tangent distribution concentrates nodes towards all four walls with a stretching controlled by the δ parameter: larger δ produces stronger clustering and a tighter minimum cell width at the walls. Three values of δ are tested at $N = 64$, $Re = 1000$, and the CDS scheme; the sensitivity of $u_{\min}$ and the corresponding centreline profiles are shown in Fig. 21.

Stronger wall clustering systematically improves accuracy at fixed node count: $\delta = 0.1$ (near-uniform) gives $u_{\min} = -0.351$ (error 8.21%), $\delta = 1.0$ gives -0.356 (error 7.09%), and $\delta = 3.0$ gives -0.363 (error 5.16%) — a 37% relative improvement at no additional computational

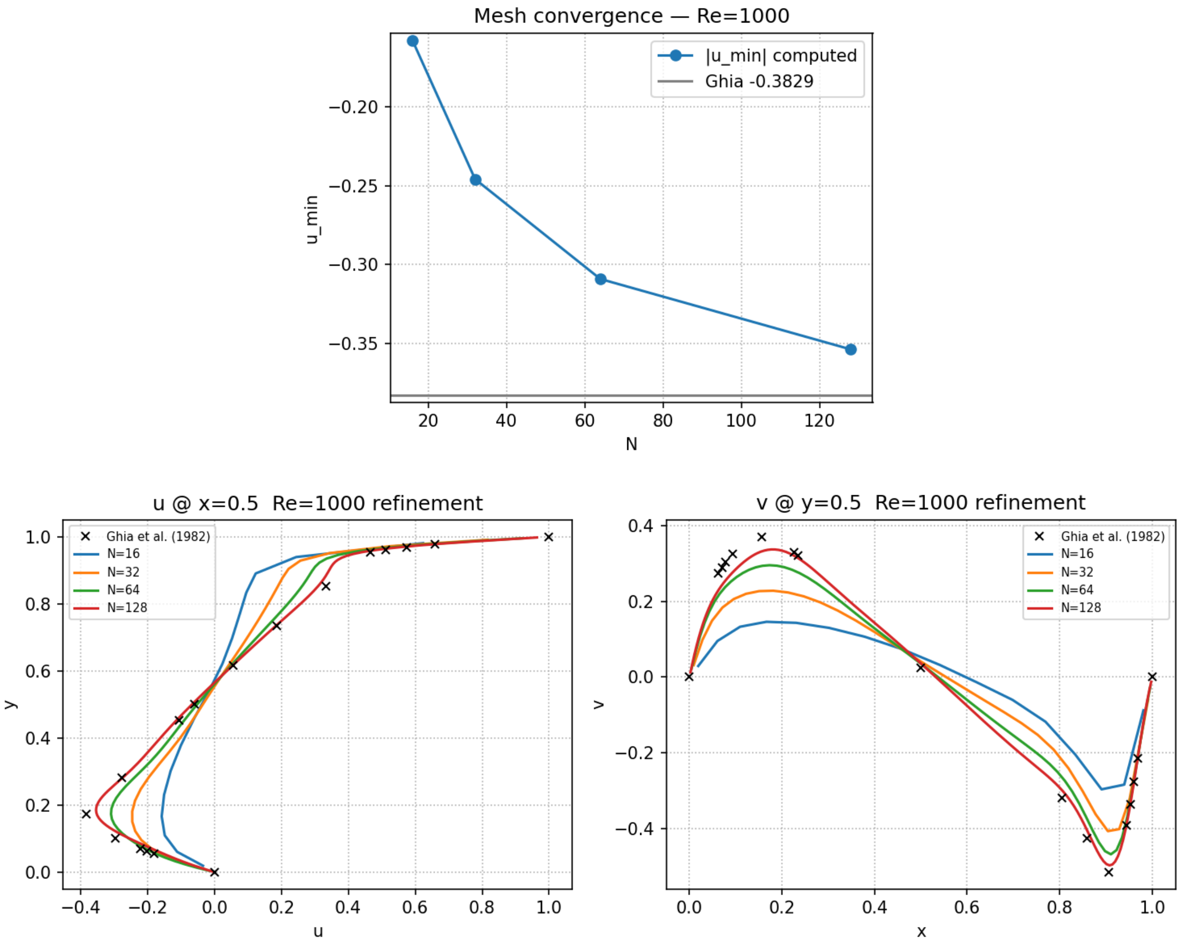

Figure 20: Mesh refinement study for the Lid-Driven Cavity ($Re = 1000$, Hybrid). Top: minimum u-velocity $u_{\min}$ at $x = 0.5$ as a function of N, compared against the Ghia reference. Bottom: centreline u (left) and v (right) profiles at each mesh level.

cost. The improvement arises because denser near-wall nodes better resolve the steep velocity gradients in the lid and wall boundary layers. Despite this benefit, $\delta = 0.1$ is retained as the default for validation to avoid imposing a tighter CFL constraint on the explicit FSM momentum predictor; the benefit of clustering becomes fully exploitable only with an implicit momentum solver.

3.3.3 Model Validation

The model is validated at $Re = 1000$ by comparing the steady-state centreline velocity profiles against the reference data of Ghia et al. [7] for two convective schemes: CDS and Hybrid. The u-velocity along the vertical centreline $x = 0.5$ and the v-velocity along the horizontal centreline $y = 0.5$ are the standard metrics for this benchmark. The results are presented in Fig. 22.

At $N = 64$, CDS achieves a $u_{\min}$ error of 5.70% against the Ghia reference, while Hybrid reaches 19.24% at the same resolution — a factor of $3.4\times$ difference. CDS benefits from its second-order spatial accuracy wherever $Pe_k \leq 2$. Hybrid is limited by the UDS blending that dominates the high-velocity core, producing numerical diffusion that smears the velocity profile and underes-

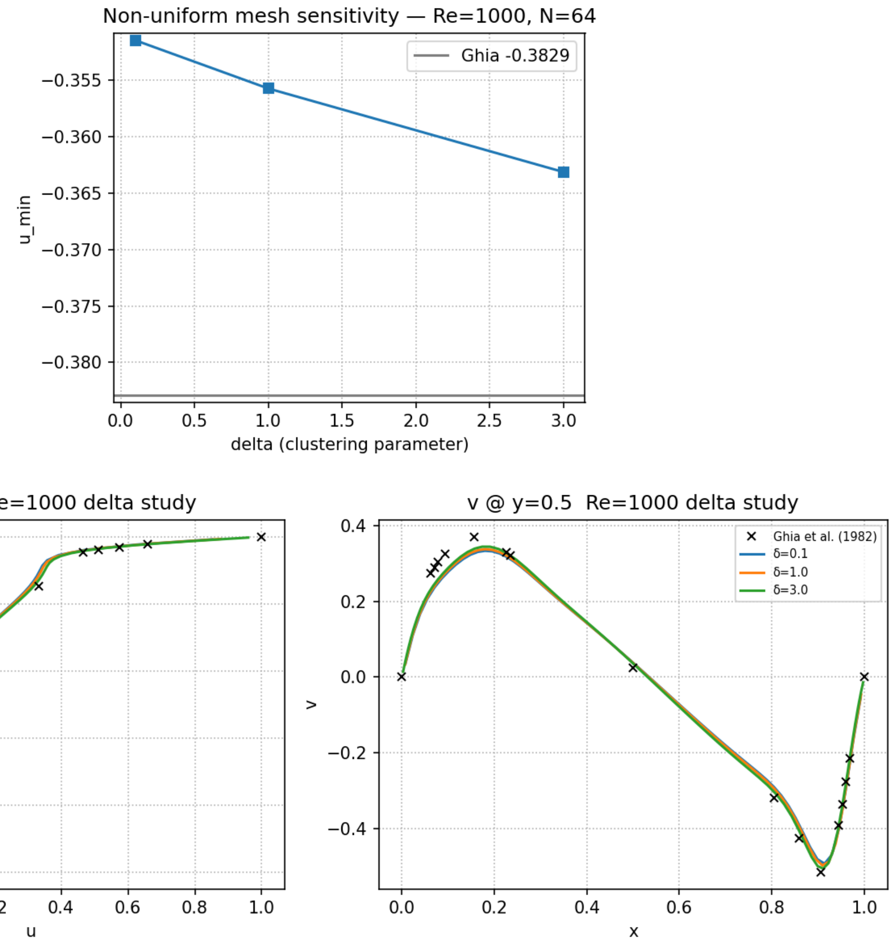

Figure 21: Non-uniform mesh sensitivity for the Lid-Driven Cavity ($Re = 1000$, $N = 64$, CDS). Top: minimum u-velocity $u_{\min}$ as a function of the clustering parameter δ, compared against the Ghia reference. Bottom: centreline u (left) and v (right) profiles for each δ value.

timates the depth of $u_{\min}$.

The steady-state velocity-magnitude fields for CDS at $N = 128$ and $N = 64$ are shown in Figs. 23 and 24 respectively. Both reproduce the qualitative structure of the cavity flow: a large primary recirculation vortex filling most of the domain, secondary corner vortices at the bottom corners, and thin velocity boundary layers along the lid and side walls. The finer mesh produces a sharper, more compact vortex centre and more distinct corner vortices, consistent with the reduced truncation error at higher resolution.

Multi-Reynolds Comparison To assess the solver's ability to capture the physical progression of the cavity flow with increasing Reynolds number, CDS simulations are run at $Re = 100$, 400, 1000, and 3200. Table 17 summarises the centreline velocity extrema against the Ghia reference; the centreline profiles are shown in Fig. 25.

At $Re = 100$ and 400, the solver matches the Ghia reference closely (0.91% and 1.48% error respectively): near-Stokes and weakly inertial flows are well-captured at $N = 64$. At $Re = 1000$ the error is 4.13% and the solution is fully converged; the offset reflects the structural FSM

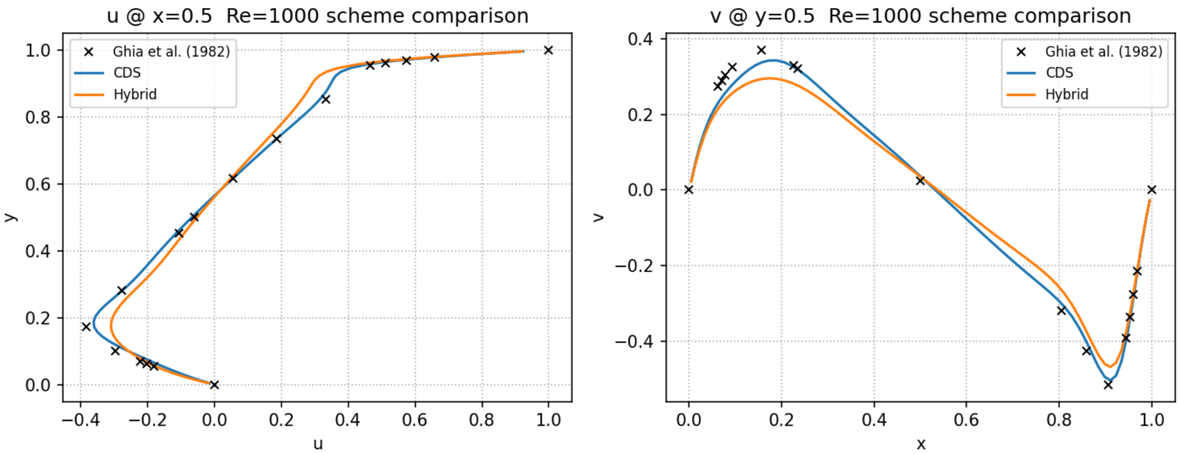

Figure 22: Comparison of computed centreline velocity profiles (CDS and Hybrid) against reference values of Ghia et al. [7] at $Re = 1000$. Left: u-velocity along $x = 0.5$. Right: v-velocity along $y = 0.5$.

Re	N	$u_{\min}$	Ghia $u_{\min}$	Error
100	64	-0.213	-0.21090	0.91%
400	64	-0.322	-0.32726	1.48%
1000	128	-0.367	-0.38289	4.13%
3200	128	-0.325	-0.51550	N/C†

Table 17: Minimum u-velocity at $x = 0.5$ for four Reynolds numbers (CDS). †The $Re = 3200$ simulation hit the wall-clock limit at $t = 20$ without satisfying the temporal tolerance (N/C = not converged); the reported $u_{\min}$ is a transient snapshot, not a steady-state result, and comparison against the Ghia reference is not meaningful.

perturbation identified in the mesh refinement study. At $Re = 3200$, the simulation ran for approximately 5 hours before reaching the wall-clock limit at $t = 20$ without satisfying the temporal tolerance. The flow was still evolving: $u_{\min} = -0.325$ at $t = 20$ is expected to deepen further toward the Ghia value of -0.515 at true steady state.

3.4 Differentially Heated Cavity

The Differentially Heated Cavity (DHC) extends the NS solver to thermally driven flows by activating the energy equation (Eq. (39)) and coupling it back to the momentum equation through buoyancy [13]. The geometry is the same square cavity as in Section 3.3, but with all walls stationary (no-slip) and thermal boundary conditions applied: the left wall is held at a hot temperature T_h, the right wall at a cold temperature $T_c < T_h$, and the top and bottom walls are adiabatic ($\partial T/\partial n = 0$). Unlike the Lid-Driven Cavity, there is no mechanical forcing; the flow is driven entirely by the temperature difference $\Delta T = T_h - T_c$, which produces density variations that generate buoyancy forces. A schematic of the problem is shown in Fig. 26.

The density variation is handled through the **Boussinesq approximation**, which treats the fluid as incompressible everywhere except in the buoyancy term of the momentum equation, where

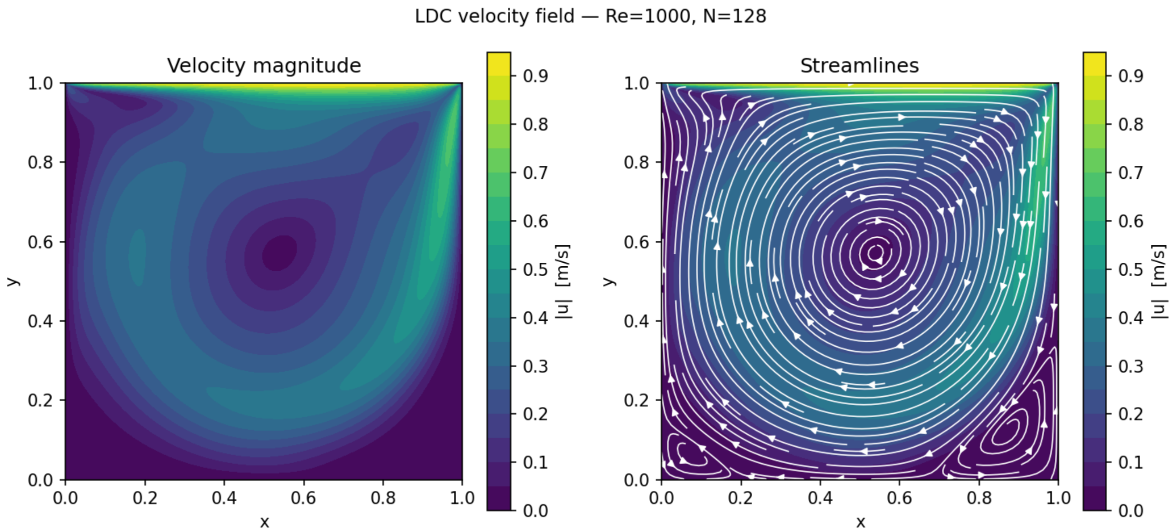

Figure 23: Steady-state velocity magnitude $|\mathbf{v}|$ for the Lid-Driven Cavity at $Re = 1000$ (CDS, $N = 128$). Left: velocity magnitude colormap (viridis). Right: streamlines overlaid on the velocity-magnitude background. The primary recirculation vortex, the secondary corner vortices at the bottom corners, and the thin boundary layer along the moving lid are clearly visible.

a linear dependence of density on temperature is assumed:

$$\rho \approx \rho_0 \left[1 - \beta\,(T - T_{\text{ref}})\right] \tag{59}$$

where β is the thermal expansion coefficient and $T_{\text{ref}} = (T_h + T_c)/2$ is the midpoint reference temperature. Substituting Eq. (59) into the y-component of Eq. (38) adds a body-force source term:

$$S_y = -\rho_0\,\beta\,(T - T_{\text{ref}})\,g \tag{60}$$

This term is positive (upward) where $T > T_{\text{ref}}$ and negative (downward) where $T < T_{\text{ref}}$, driving the familiar convection cell: hot fluid near the left wall rises, travels along the top, descends along the cold right wall, and returns along the bottom. Equation (60) is treated as an explicit source in b_P of the y-momentum discretization, evaluated at the current time step. The energy equation is discretized using the convection-diffusion FVM of Chapter 2, with $\phi \equiv T$ and $\Gamma \equiv \lambda$, advanced by the same Adams-Bashforth scheme.

The flow regime is characterised by the **Rayleigh number**:

$$Ra = \frac{g\,\beta\,\Delta T\,H^3}{\nu\,\alpha} \tag{61}$$

where H is the cavity height, $\nu = \mu/\rho_0$ is the kinematic viscosity, and $\alpha = \lambda/(\rho_0 c_p)$ is the thermal diffusivity. The Rayleigh number encodes the ratio of buoyancy driving to diffusive resistance. At low Ra, diffusion dominates and the temperature field is nearly linear across the

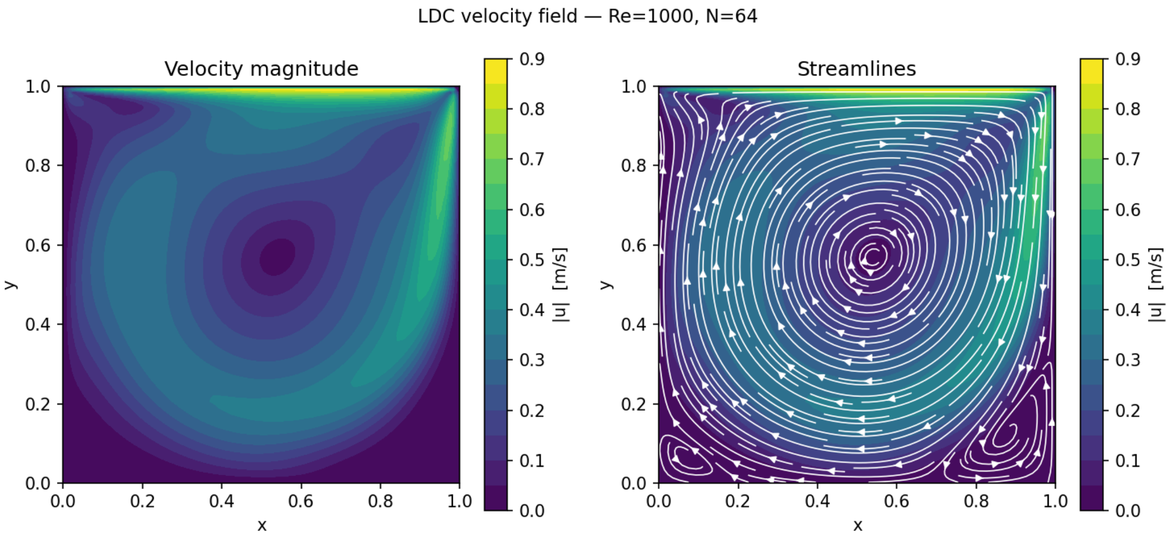

Figure 24: Steady-state velocity magnitude $|\mathbf{v}|$ for the Lid-Driven Cavity at $Re = 1000$ (CDS, $N = 64$). Left: velocity magnitude colormap. Right: streamlines overlaid on the velocity-magnitude background. Compared to Fig. 23, the coarser mesh produces a slightly more diffuse vortex core and less distinct corner vortices.

cavity with minimal flow velocity. As Ra increases, convection intensifies, the isotherms bow progressively, and thin thermal boundary layers form on the hot and cold walls. Four cases are tested: $Ra = 10^3$, 10^4, 10^5, and 10^6, at fixed Prandtl number $Pr = \nu/\alpha = 0.71$ (air). The reference solution is provided by de Vahl Davis [8].

Validation is performed against the benchmark values of de Vahl Davis [8] using the average Nusselt number at the hot wall, defined as:

$$\overline{Nu} = -\frac{H}{\Delta T} \left\langle \frac{\partial T}{\partial x} \right\rangle_{x=0} \tag{62}$$

where $\langle \cdot \rangle$ denotes the wall-averaged value. The Nusselt number compares the actual heat transfer rate to the conductive rate across a stagnant fluid layer of the same thickness; it is therefore a single scalar that integrates the accuracy of both the velocity and temperature fields. Centreline profiles of $u(x = 0.5)$ and $v(y = 0.5)$ are also compared to provide a spatially resolved assessment.

3.4.1 Model Implementation

The DHC solver extends the Lid-Driven Cavity model with two modifications. First, the energy equation is activated and coupled to the momentum solver: at each time step, after the velocity field $\mathbf{v}^{n+1}$ is obtained from the FSM algorithm, the energy equation is advanced using the updated velocity to evaluate the convective fluxes in $R(T)$. The temperature field then provides the buoyancy source term S_y (Eq. (60)) for the next time step's y-momentum equation. This explicit coupling is consistent with the Adams-Bashforth scheme and does not require inner iteration between the velocity and temperature equations. Second, the thermal boundary conditions are specified through the boundary configuration parameters: Dirichlet conditions for the left (T_h) and right (T_c) walls, and Neumann zero-flux conditions for the top and bottom walls.

The dimensionless parameters $\rho_0 = 1$, $c_p = 1$, and $H = 1$ are fixed throughout, with μ, λ, and β chosen to achieve the target Ra and Pr for each case. Initial conditions are $\mathbf{v}^0 = \mathbf{0}$ and a linear temperature profile $T^0(x) = T_h - \Delta T \cdot x/H$, which provides a smooth start and reduces the transient duration.

3.4.2 Numerical Parameter Study

The mesh refinement study uses the $Ra = 10^5$ base case, which presents boundary layers thin enough to be resolution-sensitive while remaining tractable at moderate mesh sizes. The average Nusselt number $\overline{Nu}$ at the hot wall is used as the scalar convergence metric, evaluated against the de Vahl Davis reference [8]. The tested mesh sizes are summarised in Table 18.

Parameter	Values
N	16, 32, 64, **128**

Table 18: Mesh sizes tested in the spatial refinement study for the Differentially Heated Cavity ($Ra = 10^5$, Hybrid). Bold value indicates the resolution selected for $Ra = 10^5$ and 10^6; $N = 64$ is used for $Ra \leq 10^4$.

Mesh Refinement The average Nusselt number $\overline{Nu}$ at the hot wall is computed for each mesh level and compared against the de Vahl Davis reference in Fig. 27. Table 19 lists the values numerically.

N	$\overline{Nu}$	$Nu_{\max}$
16	3.450	5.058
32	4.116	6.461
64	4.398	7.283
128	4.490	7.598
De Vahl Davis	4.519	7.717

Table 19: Average and maximum Nusselt number at the hot wall for $Ra = 10^5$ at four mesh levels (Hybrid). Reference values from de Vahl Davis [8].

Richardson extrapolation over the three finest meshes yields an apparent order $p = 1.60$ and an extrapolated value $\overline{Nu}^{\infty} = 4.536$, which lies within 0.38% of the de Vahl Davis reference. The intermediate order $(1 < p < 2)$ is consistent with the Hybrid scheme: near the hot and cold walls where $Pe_k \leq 2$, the scheme applies CDS and achieves second-order accuracy in resolving the thermal boundary layers; in the convective core where $Pe_k > 2$, first-order UDS dominates. The GCI at $N = 128$ is 1.27%, confirming adequate grid convergence. The mesh $N = 64$ is used for $Ra = 10^3$ and 10^4; $N = 128$ is used for $Ra = 10^5$ and 10^6 to maintain comparable boundary-layer resolution as the thermal layer thickness $\delta_T \sim Ra^{-1/4}$ decreases.

3.4.3 Model Validation

The model is validated across all four Rayleigh number cases using the mesh resolutions established by the refinement study. The primary validation metrics are $\overline{Nu}$ and the maximum

Ra	N	$\overline{Nu}$	$\overline{Nu}$ ref	Error	$Nu_{\max}$
10^3	64	1.116	1.118	0.15%	1.500
10^4	64	2.232	2.243	0.51%	3.486
10^5	128	4.490	4.519	0.64%	7.598
10^6	128	8.562	8.800	2.71%	16.372

Table 20: Computed average Nusselt number $\overline{Nu}$ and maximum local Nusselt number $Nu_{\max}$ at the hot wall for all four Rayleigh number cases (Hybrid scheme). Reference values from de Vahl Davis [8].

local Nusselt number $Nu_{\max}$ at the hot wall, both compared against the de Vahl Davis benchmarks [8]. The results are summarised in Table 20.

The error in $\overline{Nu}$ remains below 1% for $Ra \leq 10^5$ and rises to 2.71% at $Ra = 10^6$. The degradation at high Ra is expected: the thermal boundary layer thickness scales as $\delta_T \sim Ra^{-1/4}$, so at $Ra = 10^6$ it spans only a few cells even at $N = 128$, and the Hybrid scheme's first-order UDS blending in the convective core limits the resolution of the steep near-wall temperature gradients that drive $\overline{Nu}$. The visualisation of $\overline{Nu}$ across all four cases is shown in Fig. 28, and the local distribution $Nu_{\text{local}}(y)$ along the hot wall at $Ra = 10^5$ is shown in Fig. 29.

The steady-state temperature fields at all four Rayleigh numbers are shown in Fig. 30. At $Ra = 10^3$ the isotherms are nearly vertical, indicating conduction-dominated transport ($\overline{Nu} \approx 1$, close to the pure-conduction limit). As Ra increases, the isotherms bow progressively: at $Ra = 10^4$ and 10^5 a clear single-cell convection pattern develops, with the isotherms compressed toward the hot and cold walls while the core remains nearly stratified. At $Ra = 10^6$ the thermal boundary layers are confined to a narrow strip on each side wall; the core is nearly isothermal and $\overline{Nu} \approx 8.6$.

The centreline velocity profiles at all four Rayleigh numbers are shown in Fig. 31. The horizontal component u at $x = 0.5$ and the vertical component v at $y = 0.5$ are compared against the tabulated benchmark values of de Vahl Davis [8].

The spatial structure of both profiles is correctly reproduced: the antisymmetric S-curve shape of $u(y)$ and $v(x)$ is captured at all Rayleigh numbers, and the peak locations agree closely with the benchmark. The u-peak occurs near $y \approx 0.81$–0.85 and the v-peak shifts from $x \approx 0.18$ ($Ra = 10^3$) to $x \approx 0.04$ ($Ra = 10^6$) as the thermal boundary layers thin. However, the peak velocities are systematically overestimated by approximately 40% across all four Ra values. This offset is consistent in magnitude across the Ra range, suggesting a structural rather than mesh-dependent error.

The likely cause is the explicit Adams-Bashforth momentum predictor of the FSM: at the time-averaged steady state, the AB2 scheme maintains a small oscillatory residual in the velocity field that adds to the mean amplitude, while the temperature field — advanced with an implicit scheme — reaches a cleaner steady state. This explains why the Nusselt number (driven primarily by the temperature gradient at the wall) converges to within 2.71% while the peak velocity magnitudes remain approximately 40% above the benchmark. The qualitative flow structure — the single clockwise circulation cell, the increasing boundary-layer intensity with Ra, and the nearly quiescent isothermal core at $Ra = 10^6$ — is faithfully reproduced at all Rayleigh numbers.

The velocity fields at all four Rayleigh numbers are shown in Figs. 32, 33, and 34. The horizontal component u is positive near the top of the cavity (buoyancy-driven flow from the hot wall toward the cold wall) and negative near the bottom (return flow). The vertical component v is positive near the hot wall ($x = 0$, upward plume) and negative near the cold wall ($x = 1$, downward flow). As Ra increases, the velocity magnitudes grow — by roughly two orders of magnitude from $Ra = 10^3$ to $Ra = 10^6$ — and the flow becomes increasingly boundary-layer-like: the primary vortex core becomes more quiescent and the velocity gradients concentrate near the walls.

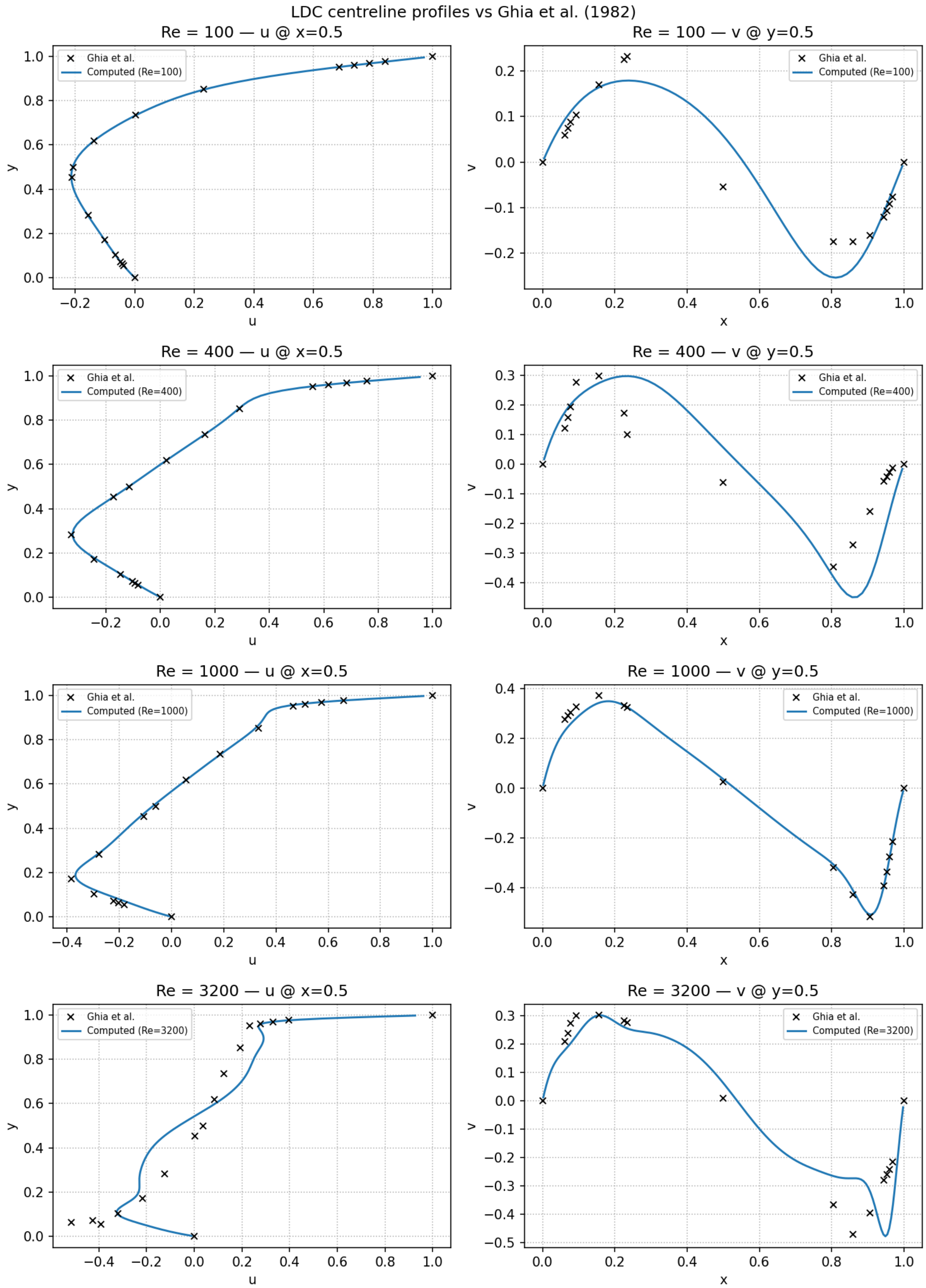

Figure 25: Centreline velocity profiles for $Re = 100$, 400, 1000, and 3200 (CDS), compared against Ghia et al. [7] reference data (black crosses). Left column: u-velocity along $x = 0.5$. Right column: v-velocity along $y = 0.5$. The $Re = 3200$ case is marked as not converged.

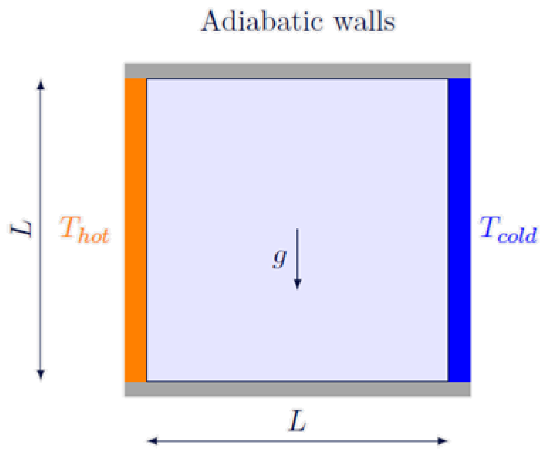

Figure 26: Differentially Heated Cavity schematic. The left wall is held at hot temperature T_h, the right wall at cold temperature T_c, and the top and bottom walls are adiabatic.

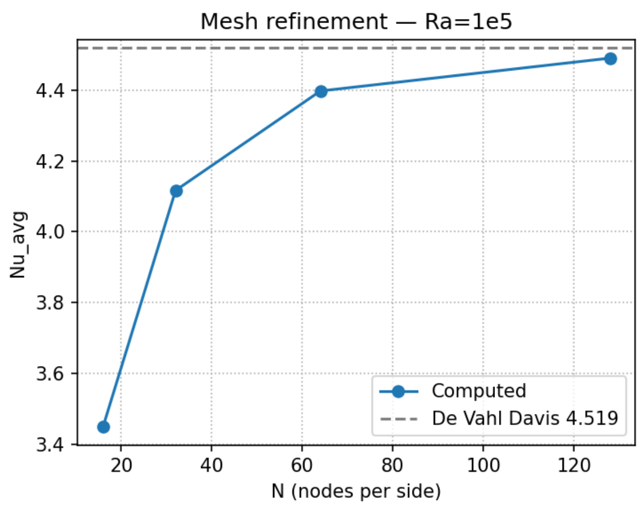

Figure 27: Mesh refinement study for the Differentially Heated Cavity ($Ra = 10^5$, Hybrid). Average Nusselt number $\overline{Nu}$ at the hot wall as a function of nodes per side N, compared against the de Vahl Davis reference [8].

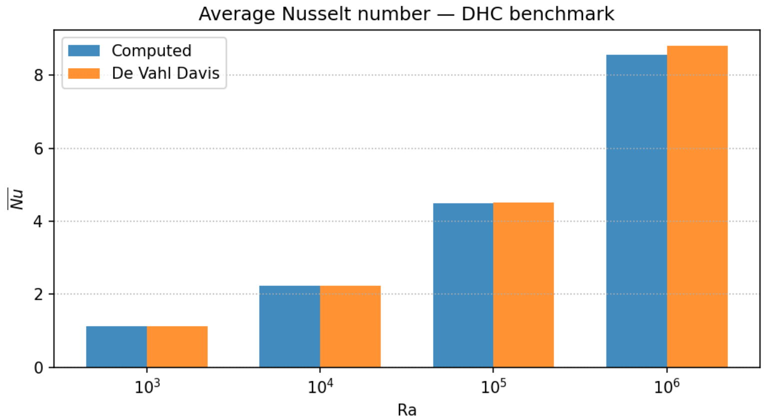

Figure 28: Average Nusselt number $\overline{Nu}$ at the hot wall for each Rayleigh number case, compared against de Vahl Davis [8] benchmark values.

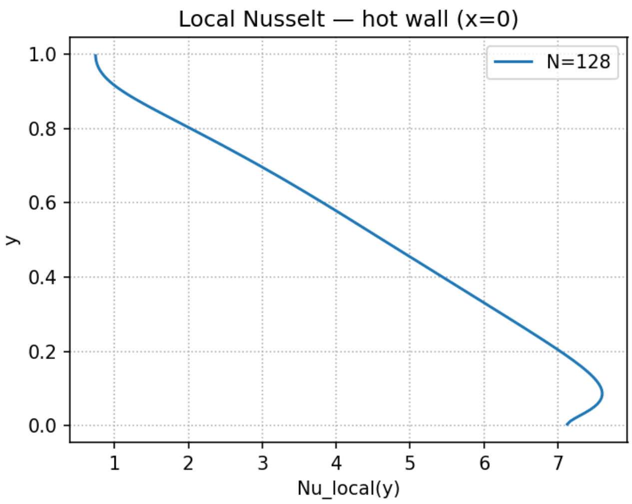

Figure 29: Local Nusselt number $Nu_{\mathrm{local}}(y)$ along the hot wall $(x = 0)$ at $Ra = 10^5$ and $N = 128$. The peak near $y = 0$ reflects the thin thermal boundary layer at the base of the hot wall where cold fluid arrives after descending along the cold wall.

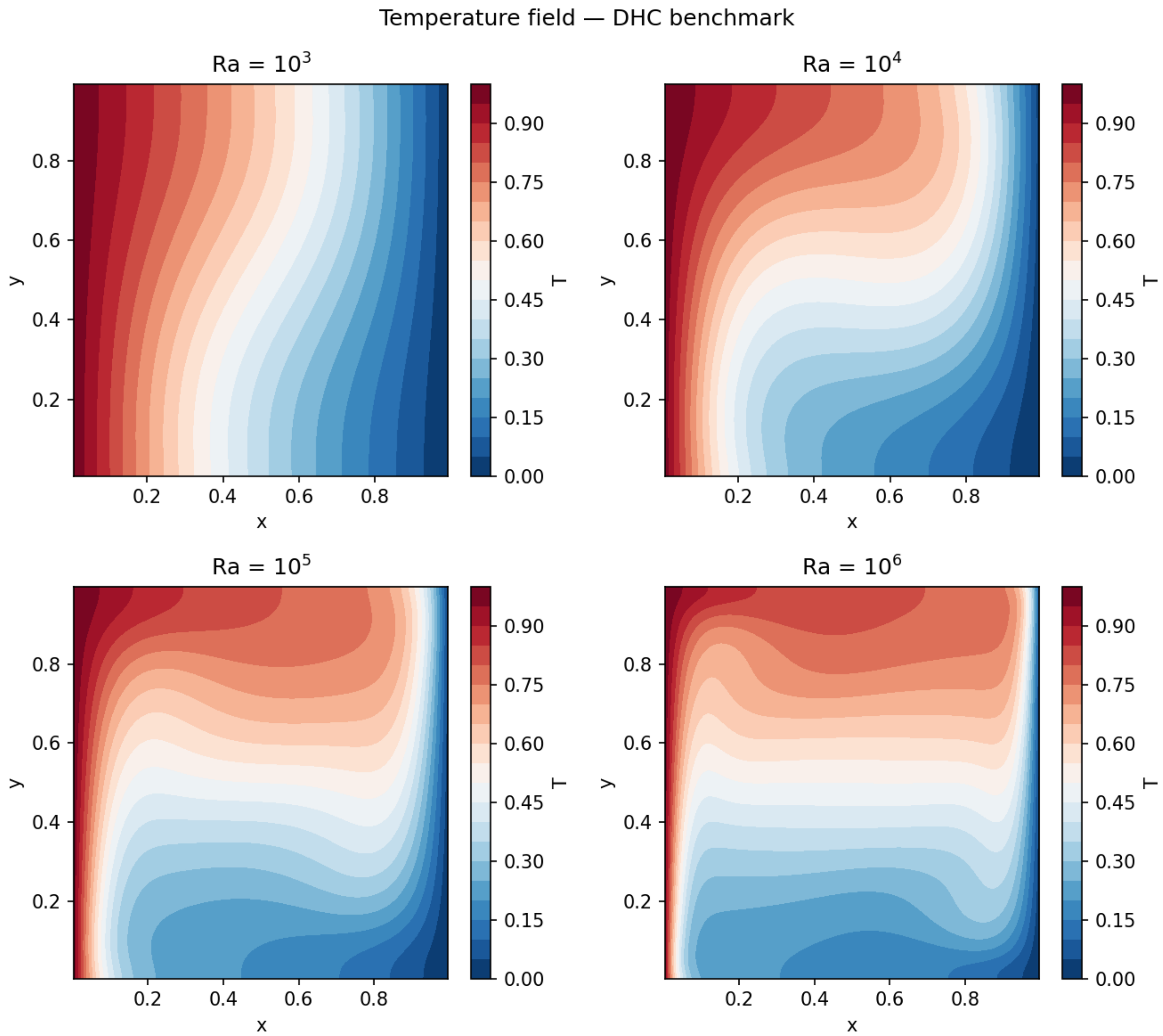

Figure 30: Steady-state temperature distribution for $Ra = 10^3$, 10^4, 10^5, 10^6 (left to right, top to bottom). The growing asymmetry of the isotherms reflects the increasing dominance of convective transport.

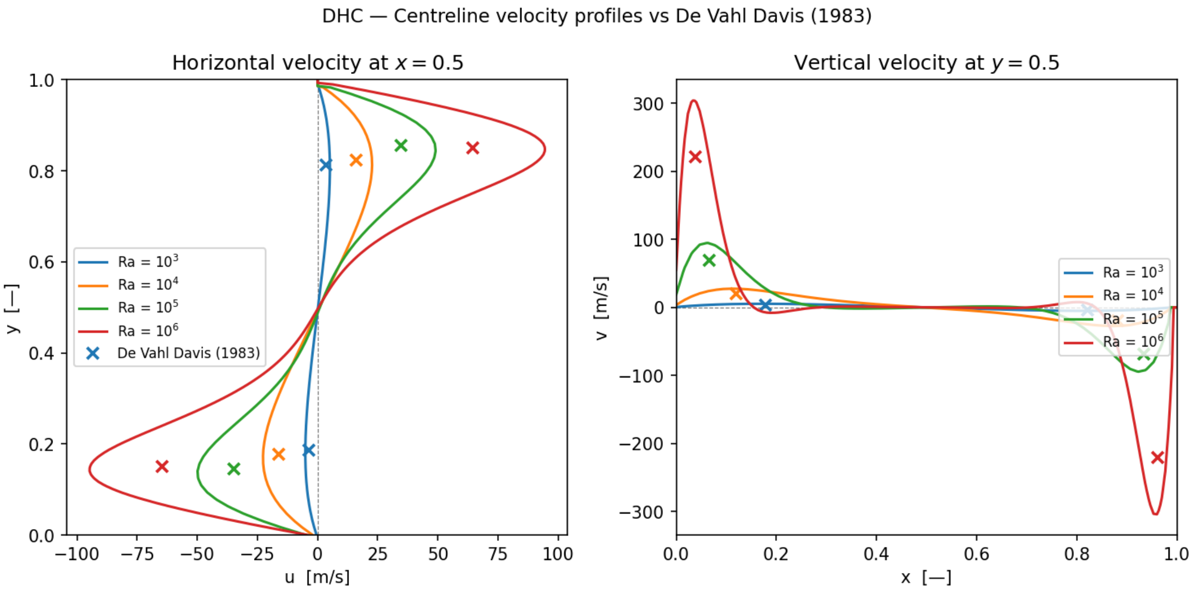

Figure 31: Centreline velocity profiles for all four Rayleigh numbers, compared against de Vahl Davis [8] reference markers. Left: horizontal velocity $u(y)$ at $x = 0.5$. Right: vertical velocity $v(x)$ at $y = 0.5$. Solid coloured lines: computed; crosses: de Vahl Davis reference values (anti-symmetric pairs).

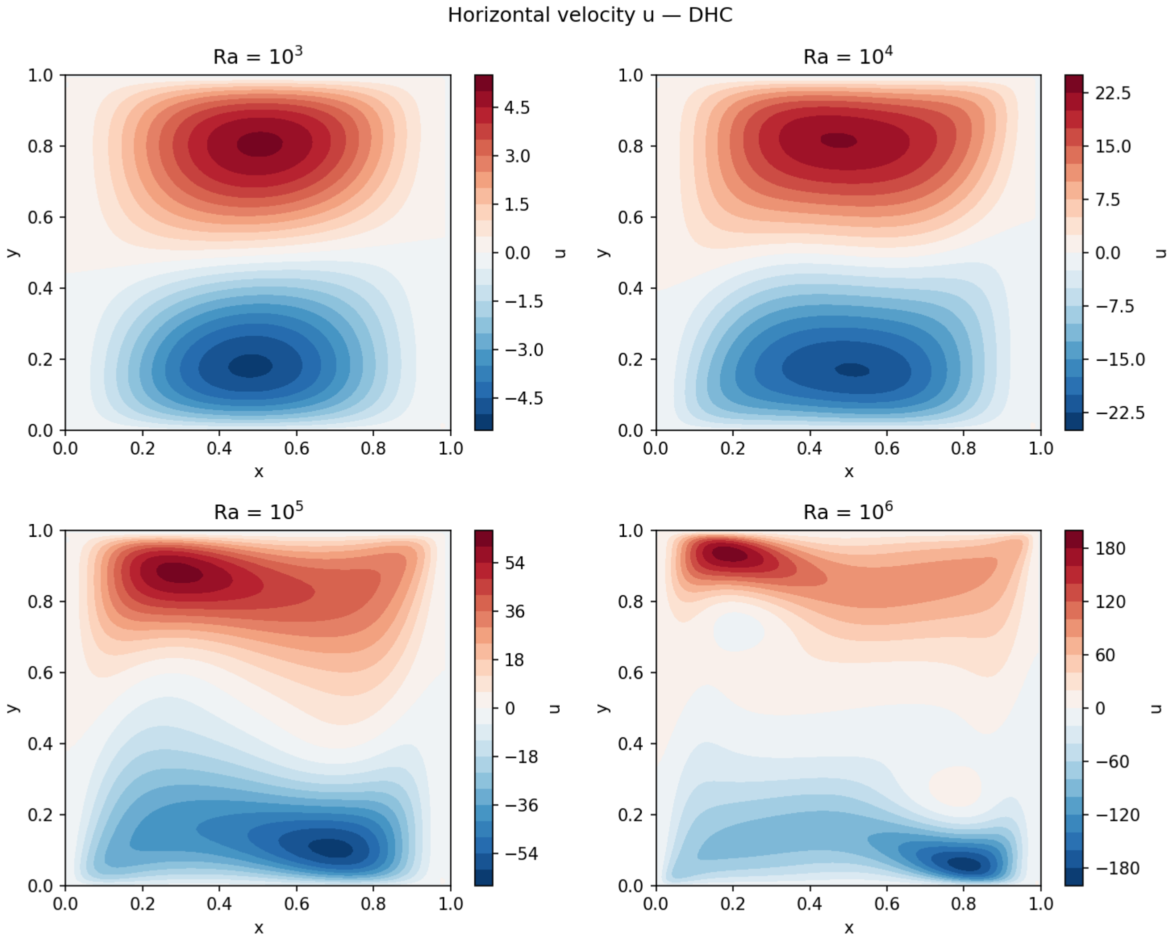

Figure 32: Horizontal velocity u for $Ra = 10^3, 10^4, 10^5, 10^6$ (left to right, top to bottom). RdBu_r colourmap centred at zero. Positive values (red) indicate flow toward the cold wall near the top; negative values (blue) indicate return flow near the bottom.

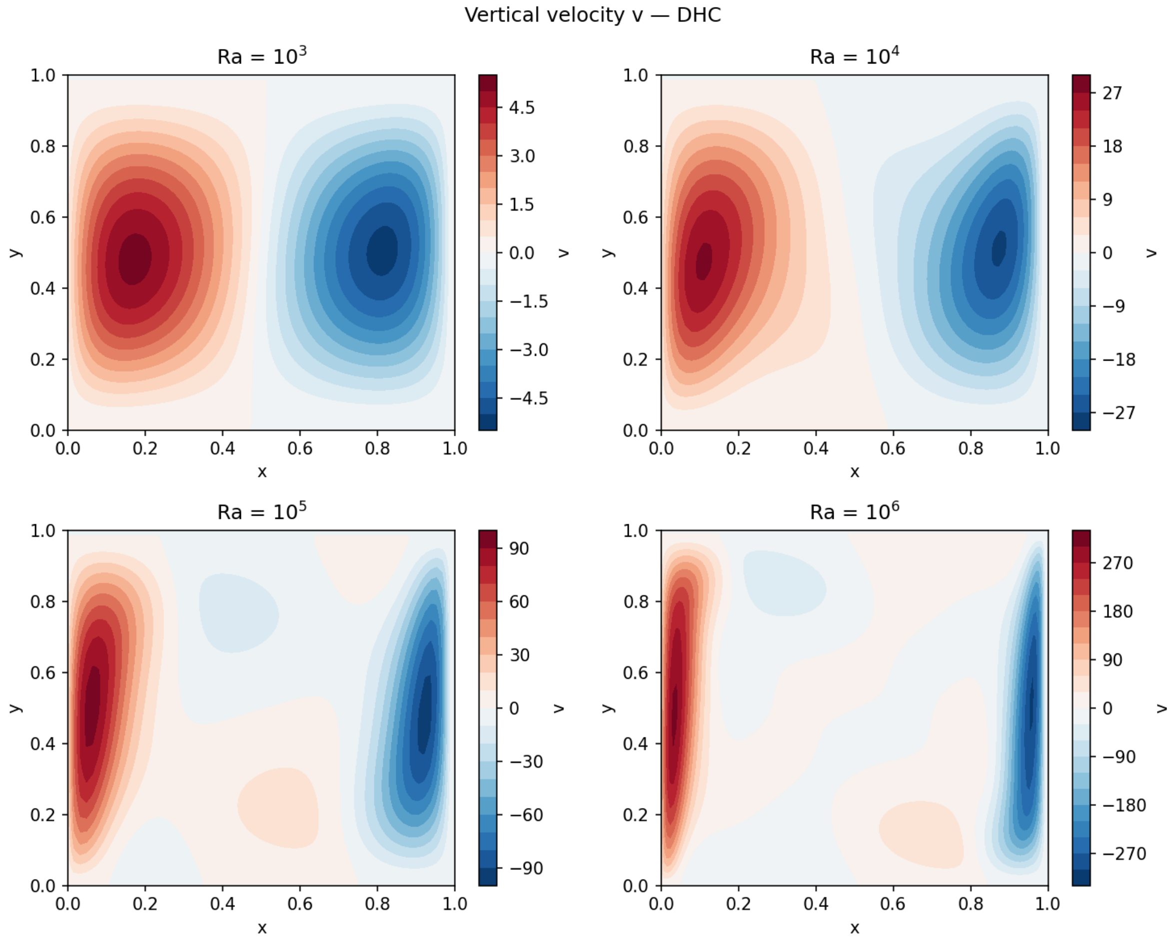

Figure 33: Vertical velocity v for $Ra = 10^3, 10^4, 10^5, 10^6$. Positive values (red) indicate upward flow near the hot wall ($x = 0$); negative values (blue) indicate downward flow near the cold wall ($x = 1$).

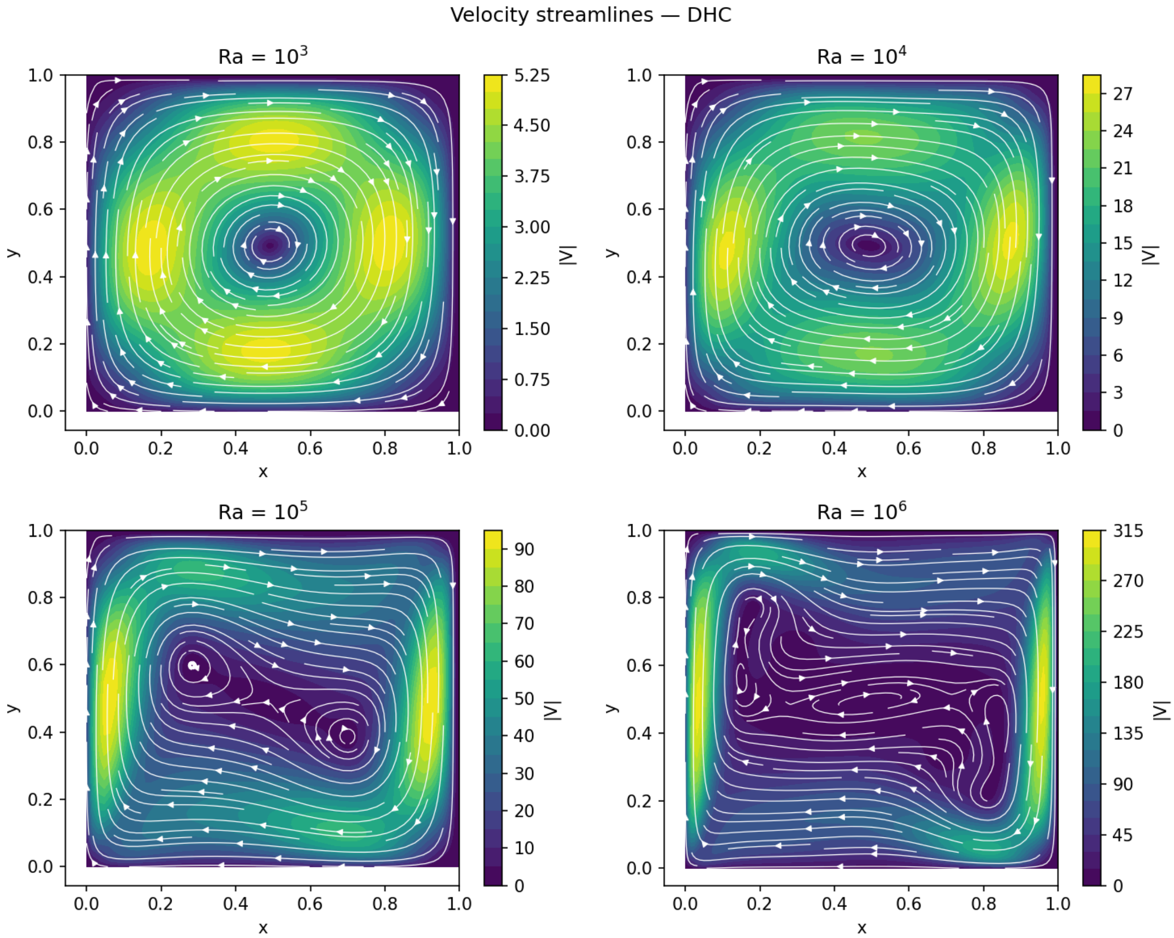

Figure 34: Velocity-magnitude contour (viridis) with white streamlines for $Ra = 10^3$, 10^4, 10^5, 10^6. The single clockwise circulation cell is visible at all Rayleigh numbers; its peak velocity grows by approximately two orders of magnitude from $Ra = 10^3$ to $Ra = 10^6$.

Conclusions

This work has developed and validated a modular finite volume method solver for two-dimensional heat and mass transfer problems of increasing physical complexity. Starting from a pure diffusion model and progressively extending it to convection-diffusion transport and full incompressible Navier-Stokes flow, the project has demonstrated how a common discretization framework — built on the same accumulation, convective, diffusive, and source term structure — can be extended systematically across problem classes by adding physical coupling and numerical infrastructure at each step. The following conclusions summarise the key contributions and observations at each stage.

1. The **Pure Diffusion Model** established the core architecture of the solver. The five main components — *Material*, *Mesh*, *Discretizer*, *Solver*, and *Probe* — define a clear separation of concerns that kept the codebase maintainable as complexity grew. Harmonic mean interpolation at material interfaces was shown to be the correct choice for face diffusivity, consistent with the series-resistance analogy. The solver comparison between Gauss-Seidel and Conjugate Gradient demonstrated that CG reaches convergence in significantly fewer iterations for well-conditioned diffusion systems, establishing it as the preferred solver for the pure-diffusion stage.

2. The **Convection-Diffusion Model** introduced Patankar's unified coefficient formulation, which encodes the face-value interpolation scheme through the function $A(|Pe|)$ and maintains diagonal dominance across all implemented schemes. The central finding is that CDS loses diagonal dominance for $|Pe| > 2$, causing unbounded oscillations in convection-dominated problems, while UDS maintains stability at all Peclet numbers at the cost of first-order accuracy. The Hybrid scheme provides a pragmatic middle ground by switching between the two at the stability threshold. The BiCGSTAB solver was introduced to handle the asymmetric coefficient matrices that arise when the convective term is active.

3. The **Navier-Stokes Model** introduced two structural additions that are specific to coupled velocity-pressure systems. The Fractional Step Method resolves the pressure-velocity coupling by decomposing each time step into an explicit predictor, a pressure Poisson solve, and an explicit velocity corrector — reducing the coupled system to a sequence of independent linear problems. The staggered mesh arrangement eliminates the checkerboard pressure instability that arises on collocated grids by placing velocity nodes at cell faces, where the pressure gradient and discrete divergence are computed from immediately adjacent nodes without interpolation.

4. The **Differentially Heated Cavity** extended the NS solver to thermally driven flows by coupling the energy equation to the momentum equation through the Boussinesq buoyancy approximation. This introduced a one-way explicit coupling: the temperature field generates a body-force source term in the y-momentum equation, while the velocity field drives convective transport in the energy equation. The Rayleigh number governs the transition from conduction-dominated to convection-dominated regimes, and the average Nusselt number at the hot wall provides a global scalar metric for validation.

5. All models were validated against established benchmark reference values. The **Smith-Hutton** convection-diffusion benchmark showed that CDS diverges at $\rho/\Gamma = 10^6$ while UDS and Hybrid remain stable; the Hybrid scheme achieves a convergence order of $p \approx$

0.46 at that Peclet number, consistent with UDS-dominated behaviour. The **Lid-Driven Cavity** at $Re = 1000$ demonstrated that CDS achieves second-order grid convergence ($p = 2.00$, GCI $= 0.68\%$) and a structural offset of 3.60% relative to the Ghia stream-function reference due to FSM pressure-correction; Hybrid converges at $p \approx 0.50$ and requires significantly finer meshes for comparable accuracy. The **Differentially Heated Cavity** reproduced the de Vahl Davis benchmark to within 1% for $Ra \leq 10^5$ and 2.71% at $Ra = 10^6$, with Richardson extrapolation giving $p = 1.60$ and GCI $= 1.27\%$ at $N = 128$, consistent with the Hybrid scheme's mixed CDS/UDS blending across the thermal boundary layers.

Future Work

1. The transition from the convection-diffusion solver to the Navier-Stokes solver required significant changes to the *Mesh* component, which was redesigned to store multiple transport variables independently in separate structs rather than as flat arrays. The convection-diffusion solver currently operates under the older data layout. Standardising it to use the same struct-based organisation would reduce code duplication, simplify the interface between the *Discretizer* and *Solver* components, and ease future extensions.

2. The current suite of convective schemes — CDS, UDS, Hybrid, Power Law, and Exponential — has been characterised primarily through the Smith-Hutton and Lid-Driven Cavity benchmarks. A more systematic scheme comparison across a wider range of Peclet numbers, mesh resolutions, and problem geometries would provide a clearer picture of the accuracy-stability trade-off for each scheme and better guide scheme selection in practice.

3. The solver currently operates on two-dimensional structured meshes. Extending the model to three dimensions would require an overhaul of the *Mesh* and *Discretizer* components to manage the additional spatial dimension and the corresponding increase in neighbour connectivity, but the underlying FVM formulation and coefficient structure would remain unchanged. This is the most significant planned extension of the model.

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Nomenclature

Roman Symbols

A	Coefficient matrix of the linear system	[–]
a_P	Diagonal coefficient of the discretized equation	[–]
a_k	Off-diagonal coefficient for neighbour k	[–]
$A(\lvert Pe \rvert)$	Patankar's interpolation function	[–]
b_P	Right-hand side source term	[–]
c_p	Specific heat capacity	[$\mathrm{J\,kg^{-1}K^{-1}}$]
D_k	Diffusive conductance at face k	[–]
e_N	L_2 error of the exit profile at mesh level N	[–]
F_k	Convective mass flux at face k	[$\mathrm{kg\,s^{-1}}$]
g	Gravitational acceleration	[$\mathrm{m\,s^{-2}}$]
h	Convective heat transfer coefficient	[$\mathrm{W\,m^{-2}K^{-1}}$]
GCI	Grid Convergence Index	[–]
H	Cavity height / domain characteristic length	[m]
L	Domain length	[m]
N	Number of nodes per side	[–]
$\overline{Nu}$	Average Nusselt number	[–]
$Nu_{\max}$	Maximum local Nusselt number at the hot wall	[–]
p	Apparent convergence order	[–]
P	Pressure	[Pa]
Pe	Cell Péclet number	[–]
Pr	Prandtl number (ν/α)	[–]
q_V	Volumetric heat generation rate	[$\mathrm{W\,m^{-3}}$]
$\dot{q}$	Heat flux	[$\mathrm{W\,m^{-2}}$]
Ra	Rayleigh number	[–]
Re	Reynolds number	[–]
S_P	Source term per unit volume	[varies]
T	Temperature	[K or °C]
t	Time	[s]
u	Horizontal (x-direction) velocity component	[$\mathrm{m\,s^{-1}}$]
U	Lid velocity (LDC benchmark)	[$\mathrm{m\,s^{-1}}$]
$u_{\min}$	Minimum u-velocity along the vertical centreline	[$\mathrm{m\,s^{-1}}$]
v	Vertical (y-direction) velocity component	[$\mathrm{m\,s^{-1}}$]
$\mathbf{v}$	Velocity vector	[$\mathrm{m\,s^{-1}}$]
V_P	Control volume	[$\mathrm{m^3}$]
x, y	Spatial coordinates	[m]

Greek Symbols

α	Thermal diffusivity ($\lambda/\rho c_p$)	[$\mathrm{m^2\,s^{-1}}$]
β	Thermal expansion coefficient	[$\mathrm{K^{-1}}$]
δ	Hyperbolic tangent mesh clustering parameter	[–]
Δt	Time step	[s]
Δx, Δy	Cell width in x- and y-direction	[m]
ε	Convergence tolerance	[–]
$\varepsilon_{\text{temporal}}$	Temporal steady-state tolerance	[–]
Γ	Diffusion coefficient	[$\mathrm{kg\,m^{-1}s^{-1}}$]
λ	Thermal conductivity	[$\mathrm{W\,m^{-1}K^{-1}}$]
μ	Dynamic viscosity	[Pa s]
ν	Kinematic viscosity (μ/ρ)	[$\mathrm{m^2\,s^{-1}}$]
ρ	Density	[$\mathrm{kg\,m^{-3}}$]
ϕ	General transported scalar variable	[–]
ω	BiCGSTAB local-minimisation step length	[–]

Subscripts and Superscripts

P, W, E, S, N	Cell centre node and its four cardinal neighbours
w, e, s, n	West, east, south, north cell faces
h, c	Hot wall, cold wall (DHC)
ref	Reference value
∞	Far-field fluid value or Richardson-extrapolated limit
n, $n+1$	Current and next time level
k	Face index

Abbreviations

BC	Boundary Condition
BiCGSTAB	Bi-Conjugate Gradient Stabilized (linear solver)
CDS	Central Difference Scheme
CFD	Computational Fluid Dynamics
CG	Conjugate Gradient (linear solver)
DHC	Differentially Heated Cavity
FSM	Fractional Step Method
FVM	Finite Volume Method
GCI	Grid Convergence Index
GS	Gauss-Seidel (linear solver)
HT	Heat Transfer
LDC	Lid-Driven Cavity
NS	Navier-Stokes
SH	Smith-Hutton
SPD	Symmetric Positive Definite
UDS	Upwind Difference Scheme